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Sensory adaptation supports flexible evidence accumulation during perceptual decision making

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eLife Assessment

This **important** study measures single-unit activity in the middle temporal area (MT) of awake-behaving monkeys to test the idea that sensory adaptation contributes to flexible evidence accumulation during decision-making. **Solid** evidence is provided, showing that adaptation to different temporal contexts shapes both perceptual judgements and neural responses, but analyses aimed at establishing a direct link between them are less persuasive. This work has the potential to be of interest to a broad range of researchers working on visual perception, plasticity, and decision making.

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Abstract

Effective decision making in dynamic environments requires flexible evidence accumulation. Although models often express this flexibility as an adaptive “leak” parameter governing accumulator dynamics, its implementation in the brain may involve adaptive mechanisms operating at other stages of the decision process. We tested whether such mechanisms include adjustments in evidence encoding. We recorded single-unit activity in the middle temporal area (MT) while monkeys performed a modified random-dot motion direction-discrimination task in which an adapting stimulus with varied temporal stability preceded a behaviorally relevant test stimulus. Monkeys flexibly adjusted their decision-making behavior in a manner consistent with an adaptive leak that depended on temporal-context stability. Behavioral adjustments were reflected in context-dependent differences in sensory adaptation in MT that were independent of changes in pupil-linked arousal. These findings identify a novel role for stimulus-specific sensory adaptation in shaping the evidence available for perceptual decisions to support flexible, context-dependent evidence accumulation.

Introduction

Perceptual decisions often require the accumulation of uncertain sensory evidence over time. This accumulation process is complicated by the fact that the nature and source of incoming evidence can change even while accumulation is underway. Under these changing conditions, effective decision makers should adjust how sensory information is accumulated so that their perceptual judgments reflect current conditions rather than outdated evidence gathered before a change occurred (Gold & Stocker, 2017 ). One way to achieve this flexibility is by adopting a “leak” in the evidence-accumulation process (Usher & McClelland, 2001 ). By scaling the rate of this leak with environmental instability, previously accumulated evidence gradually loses influence in accordance with its expected obsolescence. This adaptive, leaky evidence accumulation is a key feature of decision-making behavior in dynamic environments (Glaze et al., 2015 ; Murphy et al.,

2021 [Piet et al., 2018](#)) and is often modeled using a single parameter that controls the temporal dynamics of an accumulator (Glaze et al., 2015 [Usher & McClelland, 2001](#) [Veliz-Cuba et al., 2016](#), Fig. 1).

However, unlike these simple models, the brain contains many mechanisms that could, in principle, shape the temporal dynamics of evidence accumulation. Here, we considered a potential role for stimulus-specific sensory adaptation of individual neurons encoding the evidence that is accumulated downstream to form the decision (Fig. 1). Adaptation is a hallmark of sensory systems, occurring across modalities (Dalton, 2000 [Lampl & Katz, 2017](#) [Laughlin, 1989](#) [Ulanovsky et al., 2004](#)) and species (Brenner et al., 2000 [Clarke et al., 2015](#) [Lesica et al., 2007](#) [Nagel & Doupe, 2006](#) [Tolias et al., 2001](#)). Recent theoretical work has framed adaptation as a form of active inference, whereby sensory systems not only modulate their responses to match the statistical structure of incoming signals, but also infer when those statistics have changed and how rapidly they are expected to vary (DeWeese & Zador, 1998 [Młynarski & Hermundstad, 2021](#) [Solomon & Kohn, 2014](#) [Wark et al., 2007](#) [Weber et al., 2019](#)). Consistent with this idea, recordings from isolated retinal ganglion cells have shown that the temporal dynamics of recent stimulus history can modulate adaptation (Wark et al., 2009). Yet it remains unknown whether individual neurons in primate sensory cortex undergo similar temporal context-dependent adaptation, and whether such adaptation contributes to flexible, leaky evidence accumulation for perceptual decision making.

To test this, we leveraged decisions about visual motion. These decisions depend on neurons in the middle temporal area (MT) of extrastriate visual cortex (K. Britten et al., 1992 [K. H. Britten et al., 1996](#) [Hanks et al., 2006](#) [Newsome & Pare, 1988](#) [Salzman et al., 1990](#)), which exhibit robust stimulus-specific sensory adaptation (Kohn & Movshon, 2003 [Van Wezel & Britten, 2002](#)). We manipulated the stability of recent sensory experience by modifying the classic random-dot motion direction-discrimination task to include extended exposure to an adapting stimulus that reversed direction at either a low or high frequency, creating two context-stability conditions (Fig. 2A). Immediately following this adapting stimulus, monkeys viewed and reported the direction of a test stimulus, which was identical across conditions, allowing us to isolate the influence of recent stimulus statistics on both behavior and neural activity. We found that monkeys adjusted their decision-making behavior in a manner that was consistent with a context-dependent leak in evidence accumulation. These adjustments were based, in part, on differences in how the evidence being accumulated was encoded by individual MT neurons.

We also examined whether these context-dependent behavioral and neural adjustments related to simultaneous measures of pupil diameter. Non-luminance-mediated fluctuations in pupil diameter, an index of arousal, track quantities relevant to adaptive evidence accumulation, including environmental volatility and temporal expectations (Akdoğan et al., 2016 [Nassar et al., 2012](#) [Schwiedrzik & Sudmann, 2020](#)). Pupil diameter also relates to perceptual sensitivity, bias, and variability in evidence accumulation (de Gee et al., 2014 [2017](#) [Keung et al., 2019](#) [Krishnamurthy et al., 2017](#) [Murphy et al., 2014](#)). These findings imply a role for the arousal system in optimizing perceptual decision making, likely via neuromodulatory effects on neural information-processing dynamics (Aston-Jones & Cohen, 2005 [Devilbiss et al., 2006](#) [Joshi & Gold, 2020](#) [Navarra et al., 2017](#) [Waterhouse et al., 1998](#) [Wyrick & Mazzucato, 2021](#)). Arousal systems are therefore well positioned to, in principle, influence how sensory information is encoded, accumulated, or both (Fig. 1). Our results, which show a dissociation between sensory adaptation-related and arousal-related effects on evidence accumulation, provide new insights into the complex, distributed mechanisms that contribute to flexible decision making.

Results

We trained three rhesus macaques (An, Ch, and Mi) to perform a version of a random-dot motion direction-discrimination task that combined reversals in dot-motion direction (Glaze et al., 2015 [Fig. 2A](#)) with elements of an adaptation-test paradigm (Van Wezel & Britten, 2002). Each trial consisted of two sequential epochs: an adapting epoch (Epoch 1) and a test epoch (Epoch 2). During the adapting epoch (2400 ms), the random-dot motion stimulus abruptly switched direction at

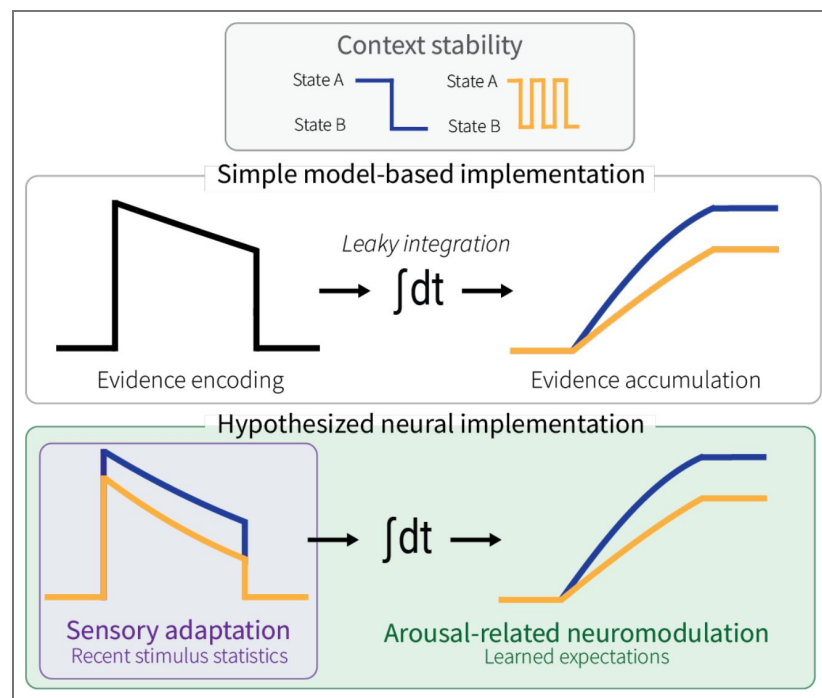


Fig. 1. Hypothesized neural implementation of a context-dependent leak in evidence accumulation.

For perceptual decisions about motion direction, momentary evidence is encoded in the middle temporal area (MT) and then accumulated by downstream circuits to form a decision variable that guides behavior. Models of decision-making often implement flexible, context-dependent evidence accumulation via a single parameter that controls the temporal dynamics (i.e., leakiness) of an accumulator (*top*). In principle, a flexible leak in the accumulation process could involve changes in evidence encoding, accumulation, or both (*bottom*). Here we used manipulations of context stability (low-versus high-frequency direction reversals of an adapting motion stimulus) to test for a role of evidence encoding. Because global signals related to arousal can influence decision flexibility and cortical information processing at multiple levels, we further considered potential interactions between adaptation- and arousal-related effects on flexible evidence accumulation

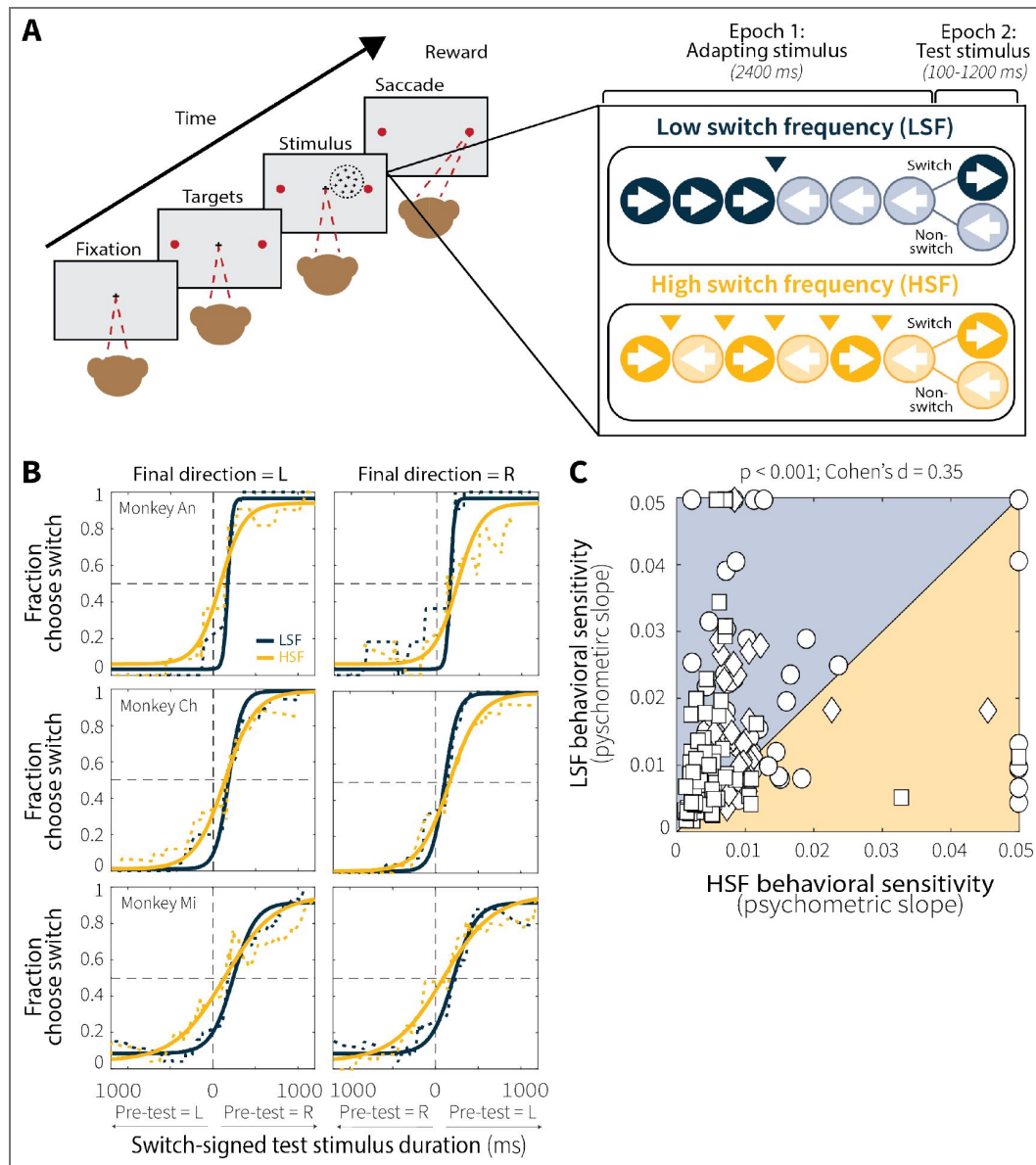


Fig. 2. Manipulating context stability in a random-dot motion task affects evidence-accumulation behavior.

(A) Each trial consisted of an adapting stimulus (2400 ms) followed immediately by a test stimulus (100–1200 ms, drawn from an exponential distribution). During the adapting stimulus, random-dot motion switched directions at either a low (LSF; 1 switch) or high (HSF; 5 switches) frequency, creating two context-stability conditions. Between epochs, there was a 50% probability of an additional change in motion direction, producing switch and non-switch trials. During the test stimulus, the monkeys accumulated motion evidence over variable durations and reported the final direction of motion with a saccade to the corresponding choice target. (B) Running average (5-trial window) of behavioral choice data (dotted line) and psychometric fits (solid lines) for representative sessions from each monkey. LSF (blue) and HSF (orange) conditions were fit separately. Shallower slopes indicate decreased perceptual sensitivity as a function of viewing time, consistent with leakier evidence accumulation. (C) Across-session comparison of fitted slopes for LSF and HSF conditions (Monkey An = 51 sessions, circles; Ch = 34, diamonds; Mi = 76, squares). p -value is from a Wilcoxon signed-rank test for equal medians.

either a low (LSF; 1 switch, occurring at 1200 ms) or high (HSF; 5 switches, occurring every 400 ms) frequency. The two adapting (“context-stability”) conditions were designed such that the total exposure time to each adapting motion direction was matched for low and high switch frequencies (1200 ms per direction). Immediately following the adapting epoch, the test stimulus was shown for a duration that was sampled randomly on each trial from a truncated exponential distribution (100–1200 ms), requiring the monkeys to accumulate evidence over varied and unpredictable timescales. At the end of this duration, the test stimulus and fixation point were extinguished simultaneously, cueing the monkeys to report the final motion direction by making a saccade to the corresponding choice target.

Critically, for both low and high switch-frequency conditions there was a 50% probability that the motion stimulus changed direction between the adapting and test epochs. This design ensured that switch and non-switch trials were equally likely, and thus that the adapting stimulus provided no information about the correct test-stimulus direction. Moreover, the transition between adapting and test epochs was marked by predictable timing and a reduction in motion coherence (to ensure that the task was difficult enough to require sustained evidence accumulation), implying that the monkeys could, in principle, learn to ignore the adapting stimulus altogether. Instead, as detailed below, systematic effects of context stability on behavior, neural activity, and pupil modulations point to inherent ways in which the brain uses recent stimulus history to shape perception.

Context stability modulates evidence-accumulation behavior

To quantify if and how the monkeys’ decisions depended on recent temporal context, we fit their choice behavior using a time-dependent logistic function (Equation 1 [↗](#)) that predicted the probability of reporting a switch in motion direction as a function of test-stimulus viewing time and trial type (switch or non-switch; Fig. 2B [↗](#)). This model captured behavior well, with Tjur’s pseudo- R^2 values that exceeded those computed by shuffling the association between test-stimulus durations and switch/non-switch trial types (Extended Data Fig. 1 [↗](#)). In general, psychometric curves were shifted rightward, consistent with a reset of the evidence accumulation process with test-stimulus onset on switch, but not non-switch trials.

The temporal statistics of the adapting stimulus systematically influenced the temporal dynamics of evidence accumulation. Psychometric slopes were shallower for high compared to low switch-frequency conditions, reflecting decreased sensitivity to incoming evidence as a function of viewing time (Fig. 2B–C [↗](#); individual animals in Extended Data Fig. 2 [↗](#)). Moreover, behavioral performance reached different asymptotes across context-stability conditions, with monkeys making more accurate choices on switch trials with long viewing durations (600–1200 ms) at low relative to high switch frequency (Wilcoxon signed-rank test for equal medians; $p = 0.02$). Both this shallower rate of rise and decreased asymptote are consistent with leakier evidence accumulation following high relative to low switch-frequency adaptation, as predicted by an adaptive “leak” that scales with environmental instability. Notably, these differences occurred despite identical test-stimulus properties, durations, and equal probabilities of a behaviorally relevant switch across conditions, demonstrating that recent temporal context can shape how sensory evidence is accumulated over time.

Context stability shapes sensory adaptation in MT

We recorded activity from 155 MT single units while the three monkeys performed the task (Monkey An = 55, Ch = 13, Mi = 87). As expected, given that stimulus properties were tailored to each neuron (see Methods), the activity of both individual units (Fig. 3A [↗](#)) and the recorded population (Fig. 3B [↗](#)) were robustly modulated by motion direction, showing strong responses to motion in each cell’s preferred direction and weak responses in the opposite, anti-preferred (“null”) direction that followed stimulus switching dynamics.

Temporal context stability systematically affected responses to preferred motion during the test epoch, with reduced activity following an adaptor that switched at a high versus low frequency (Fig. 3C [↗](#); individual animals in Extended Data Fig. 3A [↗](#)). Differences in neural activity between the two context-stability conditions were best characterized by a change in overall response

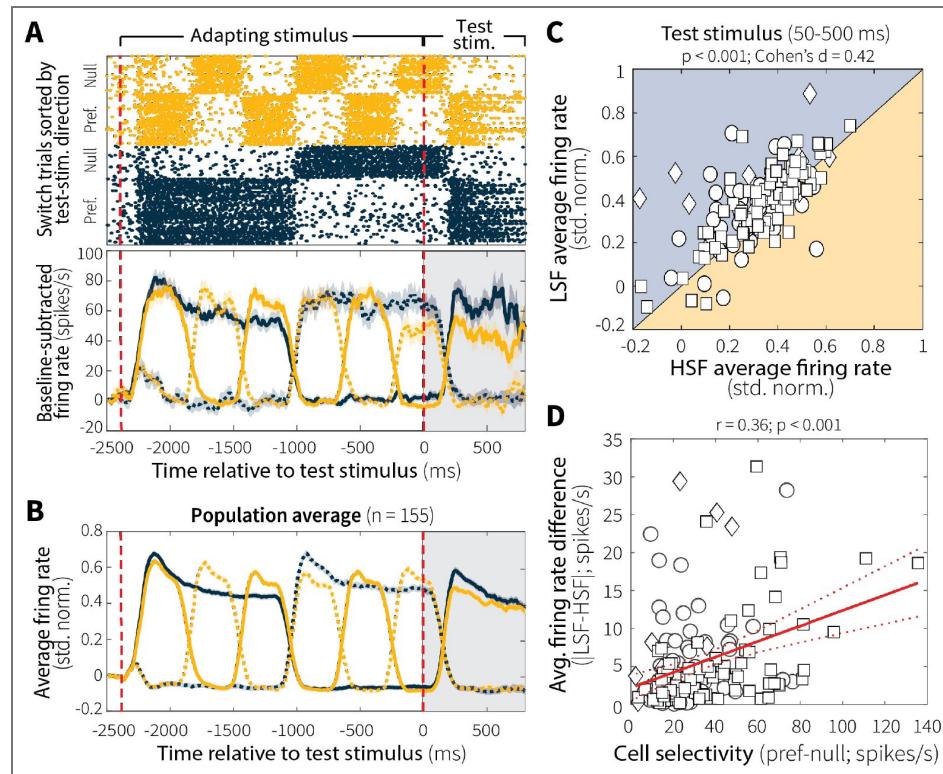


Fig. 3. Context stability modulates motion-evidence encoding in MT.

(A) Raster plot (top) and baseline-subtracted average firing rate (bottom; mean $\pm$ SEM) for switch trials from a representative MT single unit on low (LSF; blue) and high (HSF; orange) switch-frequency trials. Solid and dashed lines indicate responses to motion in the unit's preferred and null directions, respectively, defined relative to motion direction during the test stimulus. (B) Baseline-subtracted, normalized firing rate averaged across all recorded MT single units ($n = 155$; mean $\pm$ SEM). (C) Pairwise comparison of average single-unit activity during the test stimulus (50–500 ms) for LSF and HSF preferred-motion trials (Monkey An = 55 single units, circles; Ch = 13, diamonds; Mi = 87, squares). p -value is from a Wilcoxon signed-rank test for equal medians. (D) Pearson's correlation between MT direction selectivity and the magnitude of LSF–HSF response differences during the test stimulus (50–500 ms. Red solid and dashed lines indicate a linear fit $\pm$ 95% CI.

magnitude rather than a change in temporal dynamics (Extended Data Fig. 3B [↗](#)). Moreover, these differences varied systematically with direction selectivity. MT single units with stronger direction selectivity, defined as larger differences between responses to preferred and null motion, showed greater differences in activity between switch-frequency conditions (Fig. 3D [↗](#)). In other words, neurons that encoded motion evidence more strongly and more selectively were also more sensitive to the temporal statistics and stability of that evidence, exhibiting larger context stability-dependent adjustments.

These context-dependent differences in MT neural activity did not reflect baseline differences in responsiveness that persisted throughout the low and high switch-frequency blocks; rather, they emerged over the course of each trial, consistent with stimulus-specific sensory adaptation. Specifically, the average population responses to the first presentation of preferred motion during the adapting stimulus were initially matched between conditions (Fig. 4A [↗](#); although one animal showed an early difference, possibly reflecting learned expectation about switching dynamics within blocks, Extended Data Fig. 4 [↗](#)). Context-dependent differences developed with repeated stimulus presentation within a trial, culminating in robust differences in neural responses to the test stimulus between low and high switch-frequency conditions (Fig. 4A,B [↗](#)).

The effect of context stability on evidence encoding was consistent across distinct patterns of adaptation. Whereas most MT single units showed persistent response decrements with repeated preferred-motion presentations (Fig. 4C [↗](#), *bottom*), some exhibited response increments (Fig. 4C [↗](#), *top*). We classified these two groups of neurons as “adapting” and “facilitating,” respectively, based on changes in baseline-subtracted, normalized responses between the first presentation of preferred motion and preferred motion during the test stimulus (see Methods). Regardless of whether neurons were adapting or facilitating, both groups showed comparable magnitudes of response change as a function of preferred-motion stimulus presentation (Fig. 4D [↗](#)). Moreover, regardless of group, MT neurons exhibited robust context-stability effects with diminished activity at high versus low switch frequency during the test stimulus (Fig. 4E,F [↗](#)). Thus, context-specific differences in evidence encoding arose from stimulus-specific sensory adaptation that unfolded over repeated preferred-motion exposure and was expressed across MT neurons with different response profiles.

Context-dependent evidence encoding in MT relates to evidence-accumulation behavior

To assess whether context-dependent differences in MT neural responses were behaviorally relevant, we first quantified how well individual neurons discriminated between preferred and null motion at low versus high switch frequency using ROC area (Fig. 5A [↗](#)). Across the population, ROC area was lower for high versus low switch frequency (Fig. 5B [↗](#); individual animals in Extended Data Fig. 5A [↗](#)), indicating decreased evidence discriminability in MT when the test stimulus was preceded by a more unstable adapting context. We then tested whether, across sessions, these differences in ROC area predicted context-dependent behavioral differences. We found that differences in evidence discriminability (LSF–HSF ROC area, in the epoch 200–400 ms following test-stimulus onset that showed the largest context-dependent differences in neural activity) correlated with context-stability differences in behavioral performance (LSF–HSF percent correct for trials ending 375–600 ms following test-stimulus onset, Fig. 5C [↗](#); the effect was significant across animals but driven primarily by Monkey An, Extended Data Fig. 5B [↗](#)), such that greater differences in discriminability between low and high switch frequency predicted larger differences in performance.

To further explore this relationship, we divided sessions based on evidence-accumulation behavior. For sessions in which the monkeys performed leakier evidence accumulation (i.e., had shallower psychometric slopes) at high switch frequency (Fig. 5D [↗](#)), MT neurons exhibited a corresponding decrease in firing rate and discriminability (pairwise comparisons for firing rates in Extended Data Fig. 6 [↗](#)). Conversely, for sessions in which the monkeys performed leakier evidence accumulation at low switch frequency or accumulated equally across context-stability conditions (Fig. 5E [↗](#)), context-stability differences in MT evidence encoding and discriminability

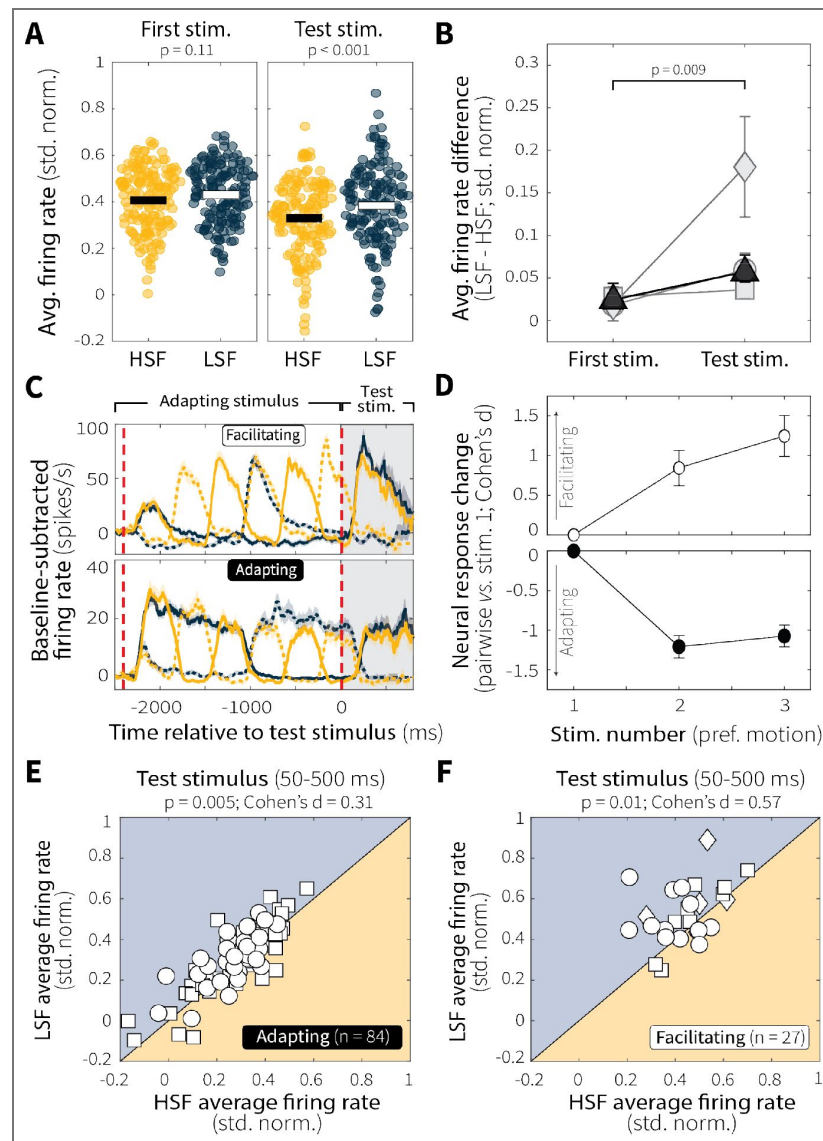


Fig. 4. Context-dependent differences in MT evidence encoding emerge following repeated preferred-motion stimulus presentation.

(A) Responses of MT single units (points) to the first (adapting-stimulus onset) and final (test) presentations of preferred-motion stimuli. Horizontal bars represent means. p -values are for two-sample t -tests for equal means of the two distributions. (B) Change in response differences (LSF-HSF) from the first (adapting-stimulus onset) and final (test) presentations of preferred-motion stimuli. Individual animals are in gray; group average is in black; all points are mean $\pm$ SEM across units. p -value is from a two-sample t -test comparing the full distribution of response differences (LSF-HSF) from the two time intervals for all monkeys. (C) Example responses from representative facilitating (top) and adapting (bottom) single units. Colors and line styles as in Fig. 2A. (D) Comparison of average baseline-subtracted and normalized responses across successive presentations of preferred motion within the adapting epoch for HSF switch trials (panel C, orange solid). For adapting (black, $n = 87$) and facilitating (white, $n = 27$) groups, responses to each preferred-motion presentation were compared to the first (adapting-stimulus onset) preferred-motion response using Cohen's d to quantify changes in response magnitude as a function of stimulus number. Points denote mean $\pm$ SEM across units. (E, F) Pairwise comparison of average single-unit responses to the test stimulus (50-500 ms) during LSF versus HSF conditions for adapting (E) and facilitating (F) units. Different symbols represent data from different monkeys, as in Fig. 2C. p -values are from a Wilcoxon signed-rank test for equal medians.

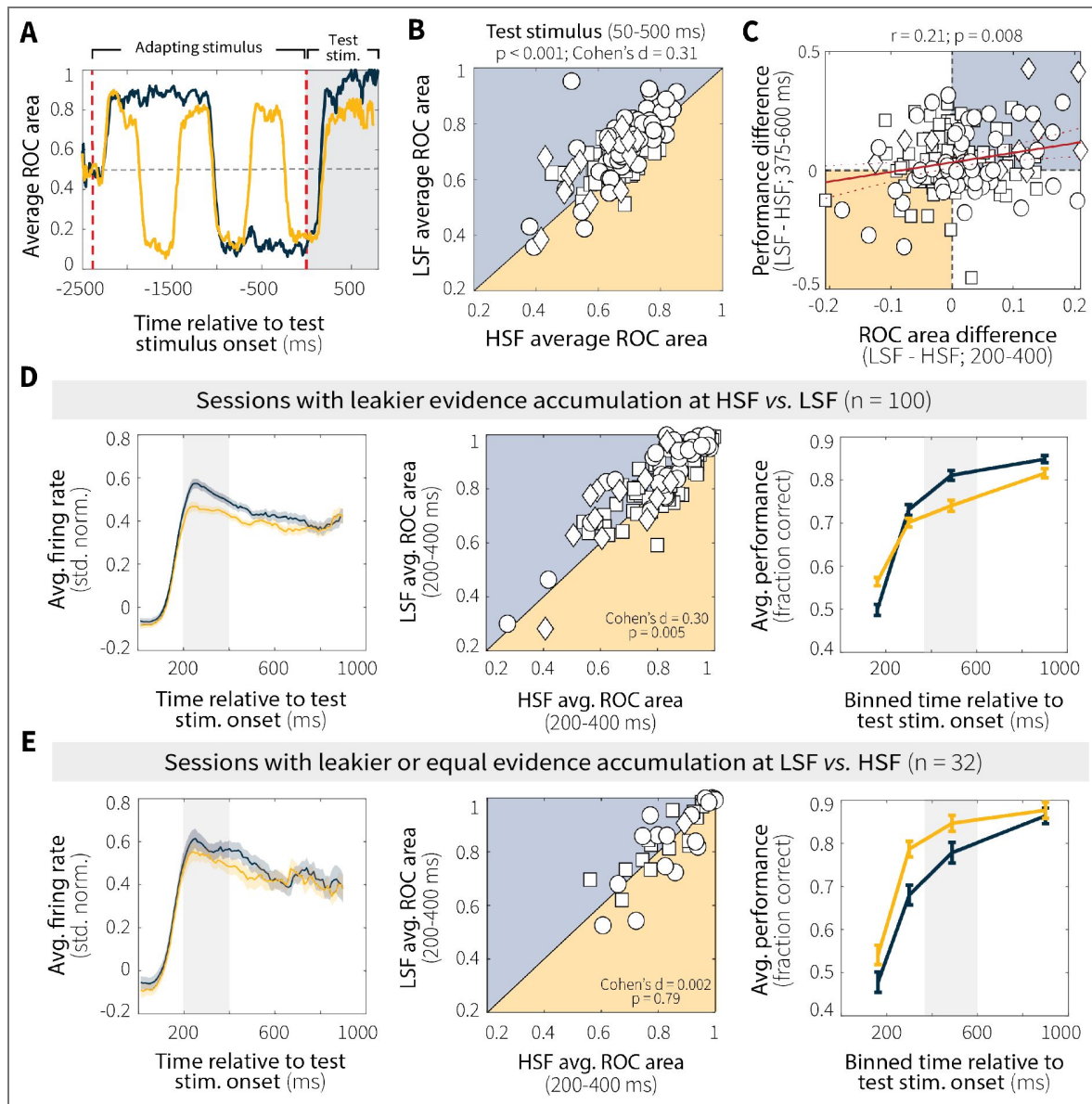


Fig. 5. MT neural activity relates to evidence-accumulation behavior.

(A) Average ROC area over time for a representative MT single unit showing discriminability between preferred and null motion during low (LSF, blue) and high (HSF, orange) switch-frequency switch trials. (B) Pairwise comparison of average ROC area during the test stimulus (50–500 ms) for LSF and HSF switch trials. p -value is from a Wilcoxon signed-rank test for equal medians. (C) Context stability-dependent differences in behavioral performance differences (LSF–HSF, measured as the difference in accuracy for trials that ended 375–600 ms after test-stimulus onset; gray band in D and E, left panels) plotted as a function of MT single-unit discriminability (LSF–HSF ROC area, 200–400 ms after test-stimulus onset; gray band in D and E, right panels). Red solid and dashed lines indicate a linear fit $\pm 95\%$ CI. p -value is for the Pearson's correlation coefficient ($H_0: r = 0$). (D) Sessions with shallower psychometric slopes at HSF relative to LSF. Columns show (left) MT evidence encoding (mean $\pm$ SEM baseline-subtracted, normalized firing rates averaged across units), (middle) MT evidence discriminability (average ROC area, 200–400 ms after test-stimulus onset; p -values from a Wilcoxon signed-rank test for equal medians), and (right) behavioral performance in stimulus-duration bins with roughly equal numbers of trials (mean $\pm$ SEM across sessions). (E) Same as D, but for sessions with shallower or equal slopes at LSF relative to HSF. Different symbols in the four scatterplots represent data from different monkeys, as in Fig. 2C.

were attenuated (pairwise comparisons for firing rates in [Extended Data Fig. 6](#)). This relationship was specific to correct trials and absent for incorrect trials, which showed no modulation based on evidence-accumulation behavior ([Extended Data Fig. 7](#)). Together, these results suggest that adaptation-driven differences in MT evidence encoding contribute, at least in part, to context-dependent evidence-accumulation behavior.

Context-dependent evoked pupil modulations relate to evidence-accumulation behavior

To probe whether processes beyond local sensory adaptation contribute to adaptive evidence accumulation, we examined both baseline and evoked pupil diameter as proxies for arousal-related modulations that have been shown to encode key features of adaptive perceptual decision making ([Cheadle et al., 2014](#); [Krishnamurthy et al., 2017](#); [Murphy et al., 2021](#)). Trial-wise baseline pupil diameter was correlated with the time required for monkeys to obtain fixation ([Extended Data Fig. 8A](#)), a relationship that was robust across sessions and animals ([Extended Data Fig. 8B](#)). However, baseline pupil diameter did not differ systematically between low and high switch-frequency conditions ([Extended Data Fig. 8C](#)), suggesting that it captured global as opposed to context-dependent fluctuations in arousal or task engagement. We therefore focused on task-evoked, as opposed to baseline, pupil responses in subsequent analyses.

Evoked pupil responses were modulated by context stability. Following a brief, stereotyped constriction caused by the abrupt luminance increase at motion-stimulus onset, differences in pupil diameter emerged between low and high switch-frequency conditions ([Fig. 6A](#)). To quantify these context-dependent effects, we used sliding-window linear regressions ([Equation 2](#)) to estimate differences in evoked pupil diameter (LSF–HSF), while controlling for baseline pupil diameter (because evoked and baseline diameter tend to be inversely related; [Mathôt et al., 2018](#)). For both monkeys (Monkey Ch was excluded from pupil-related analyses; see [Methods](#)), the effect of context-stability on evoked pupil diameter increased over the course of the adapting stimulus ([Fig. 6B](#); individual animals in [Extended Data Fig. 9A](#)). This result is consistent with the idea that evoked pupil responses track context-dependent expectations or anticipatory processes associated with behaviorally relevant stimuli.

Context-dependent differences in evoked pupil diameter were relevant to behavior. Specifically, differences in evoked pupil diameter (LSF–HSF) showed correlations with behavioral sensitivity (LSF–HSF psychometric slope) that varied over the course of the trial and peaked just before the onset of the test stimulus ([Fig. 6C](#)). Around this peak (–500–0 ms before test-stimulus onset), average context-dependent differences in evoked pupil diameter reliably predicted subsequent differences in behavioral sensitivity across sessions ([Fig. 6D](#); individual animals in [Extended Data Fig. 9B](#)). As such, sessions with larger differences in evoked pupil diameter also tended to show larger context-dependent differences in behavioral sensitivity.

Adaptation and arousal-related mechanisms are jointly and differentially recruited across sessions

To directly compare the relative contributions of MT sensory adaptation and arousal-related mechanisms to adaptive evidence-accumulation behavior, we separately incorporated trial-wise measures of MT neural activity and evoked pupil diameter into the psychometric functions ([Equation 3](#)). This approach allowed us to quantify how strongly each signal modulated behavioral sensitivity as a function of context stability (LSF–HSF) on a session-by-session basis.

Both neural and pupil terms improved model fits, as evidenced by increased explanatory power relative to the behavior-only model in both animals ([Fig. 7A](#)). However, the relative improvement attributable to neural activity versus evoked pupil diameter differed between monkeys, with Monkey An showing a larger difference between neural versus pupil terms than Monkey Mi (Wilcoxon rank-sum test on neural–pupil differences in Tjur’s pseudo- R^2 ; $p < 0.001$, [Fig. 7A](#)), implying joint, but flexible recruitment of adaptation and arousal-related mechanisms.

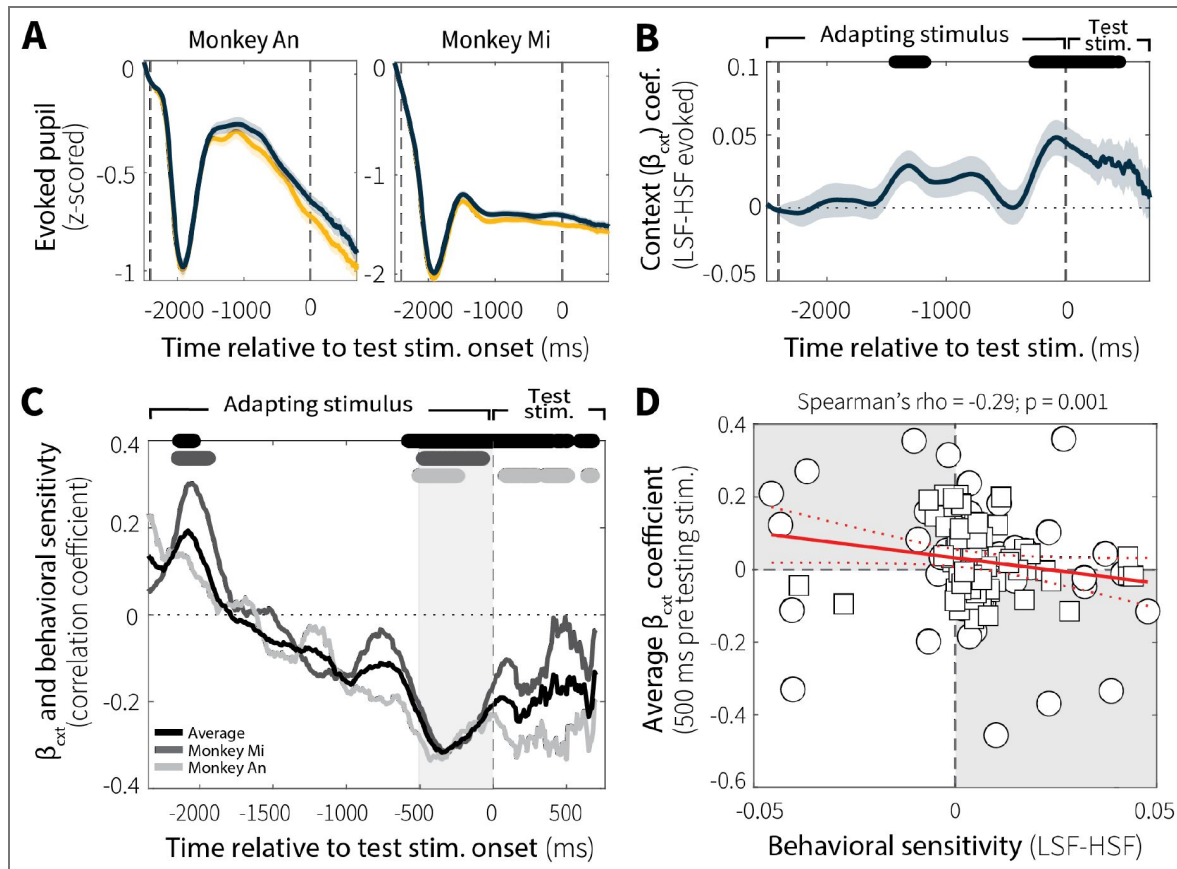


Fig. 6. Evoked pupil responses depend on context stability and relate to evidence-accumulation behavior.

(A) Average evoked pupil responses over time (z-scored per session) for low (LSF; blue) and high (HSF; orange) switch-frequency conditions (mean $\pm$ SEM across all sessions), shown separately for the two monkeys. (B) Sliding-window regression coefficients (β_{cxt} ; mean $\pm$ SEM across all sessions from both monkeys) estimating the effect of context-stability condition (LSF-HSF) on pupil diameter during stimulus viewing while controlling for baseline pupil diameter. Black bar (top) indicates windows in which β_{cxt} differed significantly from zero ($p < 0.05$, uncorrected for multiple comparisons). (C) Spearman's rank correlation coefficient between behavioral sensitivity (LSF-HSF psychometric slope) and β_{cxt} coefficients from individual sessions as a function of time relative to test-stimulus onset (computed in 100 ms bins with 10 ms steps). Data from individual animals are shown separately (Monkey An: light grey; Monkey Mi: dark grey) along with the across-animal average (black). Corresponding grayscale bars at the top indicate time points with significant correlations ($p < 0.05$, uncorrected for multiple comparisons). (D) Spearman's rank correlation between average evoked pupil diameter differences (β_{cxt} , -500–0 ms relative to test-stimulus onset; gray band in C) and behavioral sensitivity differences (LSF-HSF slope) across sessions. Red solid and dashed lines indicate a linear fit $\pm$ 95% CI. p -value is for the Spearman's correlation coefficient ($H_0: \rho = 0$).

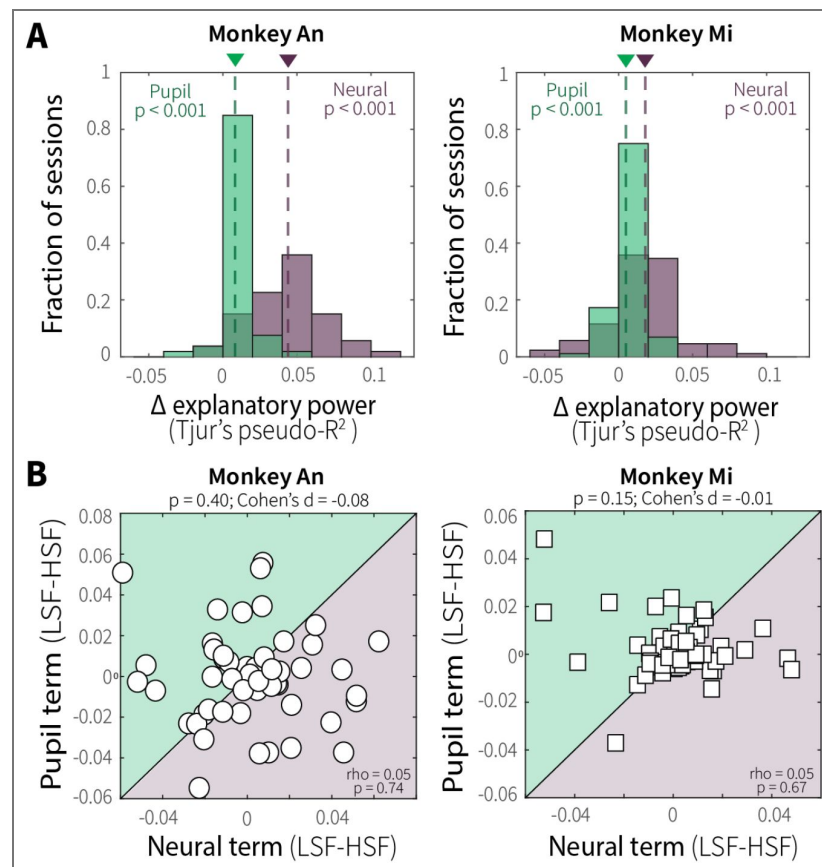


Fig. 7. Context-stability jointly and differentially recruits adaptation and arousal-related mechanisms across sessions.

(A) Distributions of differences in explanatory power (Tjur's pseudo- R^2) for models fit with versus without the neural (purple) or pupil (green) term across sessions for Monkey An (left) and Monkey Mi (right). Mean values are indicated by dashed lines and corresponding colored triangles. p -values are from a one-sample sign test for H_0 : fit parameters came from a distribution with a median of zero. (B) Context-stability differences (LSF-HSF) in neural (MT firing rate; abscissa) and pupil (evoked diameter; ordinate) model terms for Monkey An (left) and Monkey Mi (right). Points are fits to data from individual sessions/units from logistic models that quantified how much the slope of time-dependent psychometric functions covaried with the given neural or pupil term, computed separately for LSF and HSF trials. Shaded quadrants denote relative dominance of neural (purple) or pupil (green) effects. p -values are from a Wilcoxon signed-rank test for equal medians (top) and Spearman's correlation (bottom).

This flexible recruitment was evident across sessions, where the influence of context-dependent MT neural activity and evoked pupil diameter on behavioral sensitivity varied considerably, with neither term reliably predominating over the other (Fig. 7B). Session-wise, context-dependent contributions from neural and pupil terms were uncorrelated with each other, indicating that the two signals contributed separately rather than redundantly or in opposition (Fig. 7B). Consistent with this, context-dependent modulations of pupil size were not systematically related to context-dependent modulations of MT neural activity or discriminability (Extended Data Fig. 9C,D). Together, these findings suggest that sensory adaptation and pupil-linked arousal both contribute to adaptive evidence accumulation, likely operating at different processing stages, with their relative weighting varying substantially across sessions and animals.

Discussion

Flexible evidence accumulation is critical for effective decision making in dynamic environments. Motivated by algorithmic accounts proposing that this flexibility arises from an adaptive “leak” in evidence accumulation, we explored the possibility that such an apparent leak may not solely reflect properties of the accumulator itself but instead could emerge from adaptive processes operating at multiple stages of the decision process. By manipulating the temporal stability of recent sensory experience and simultaneously measuring both neural activity in the middle temporal area (MT) and pupil-linked arousal, we identify two complementary mechanisms that could support flexible evidence accumulation: one involving sensory adaptation and the other involving arousal-related neuromodulation (Fig. 1). Our results show that these mechanisms operate independently, likely at different processing stages, providing a new view of the distributed neural substrates for flexible evidence accumulation in dynamic environments.

Our MT findings imply that a behavioral signature of leakier evidence accumulation (i.e., shallower rate-of-rise of accuracy as a function of viewing time) for high versus low switch-frequency conditions is based, in part, on context-dependent differences in sensory encoding. In general, both motion-evidence encoding and discriminability during the test stimulus were reduced following more unstable adapting stimuli (Fig. 3A-C). These reductions were not sustained differences throughout the two context-stability conditions but instead emerged on each trial through repeated presentations of preferred motion at different switch frequencies, consistent with stimulus-specific sensory adaptation (Fig. 4A-B). This adaptation varied systematically across individual units, such that more direction-selective cells were more affected by differences in stimulus temporal statistics, which is consistent with previous findings that more active MT neurons adapt more strongly (Fig. 3D; Van Wezel & Britten, 2002).

It is well established that sensory adaptation can depend on stimulus statistics such as the mean, variance, and salience of sensory inputs (Dean et al., 2005; Fairhall et al., 2001) and even changes in those statistics (Kastner & Baccus, 2011; Tikiđji-Hamburyan et al., 2015). Here, we extend these findings by demonstrating that adaptation can also depend on the rate of change (i.e., temporal dynamics) of stimulus statistics. This kind of adaptation has been observed in retinal ganglion cells (Wark et al., 2009). Our work extends those findings to non-human primate cortical neurons with causal links to behavior (Hanks et al., 2006; Newsome & Pare, 1988), in particular showing that context stability-dependent adaptation in MT shapes the sensory evidence available for perceptual decisions.

We also observed heterogeneity in how MT neural activity changed over the course of the adapting stimulus, with some cells exhibiting progressive response attenuation and others facilitation (Fig. 4C-D). A plausible mechanism for this facilitation is that adaptation alters the local excitation-inhibition balance (Higley & Contreras, 2007; Solomon & Kohn, 2014). In particular, repeated presentation of null motion may weaken opponent input, producing disinhibition of preferred-motion responses (Kohn & Movshon, 2003). Consistent with this account, facilitating neurons showed increased responses to preferred motion following null-motion adaptation (Extended Data Fig. 10). This diversity of adaptation profiles mirrors observations in retinal ganglion cells (Kastner & Baccus, 2011, 2013), primary auditory cortex (Seay et al., 2020), and somatosensory cortex (Adibi et al., 2013; Cohen-Kashi Malina et al.,

2013 [↗](#); Derdikman et al., 2006 [↗](#); Kheradpezhohu et al., 2017 [↗](#)), and recent work suggests that whether a neuron adapts or facilitates may itself be a flexible property that depends on recent sensory experience (Dobler et al., 2024 [↗](#)). Nevertheless, regardless of the adaptation profile, context stability shaped MT activity and discriminability in ways that corresponded to context-dependent differences in evidence-accumulation behavior across sessions.

These adaptation-driven changes in MT evidence encoding were relevant to behavior, but not fully prescriptive of context-dependent adjustments in evidence accumulation. For instance, for sessions in which the monkeys showed behavioral evidence of leaky, context-dependent evidence-accumulation (i.e., reduced sensitivity as a function of viewing time at high relative to low switch frequency; Fig. 5D [↗](#)), there was a corresponding change in MT evidence encoding (e.g., reduced neural activity at high relative to low switch frequency; Fig. 5D [↗](#)). However, sessions showing the opposite pattern of behavior did not show opposite neural effects (Fig. 5E [↗](#)). Thus, while sensory adaptation supports flexible evidence accumulation by shaping the fidelity of sensory input, additional mechanisms likely contribute to how that information is dynamically weighed and accumulated over time.

Consistent with this idea, fluctuations in task-evoked pupil size, an index of arousal-related neuromodulation, were also systematically related to temporal context stability. Differences in evoked pupil diameter emerged between low and high switch-frequency conditions over the course of the adapting epoch, peaking just before test-stimulus onset (Fig. 6B [↗](#)). This time course suggests that evoked pupil dynamics may be tracking expectations about environmental statistics to prepare downstream decision circuits for the behaviorally relevant stimulus, consistent with previous work (Filipowicz et al., 2020 [↗](#)). Across sessions, the magnitude of context-dependent differences in evoked pupil diameter predicted corresponding differences in behavioral sensitivity (Fig. 6C-D [↗](#)). Specifically, larger pupil diameter corresponded to leakier evidence accumulation, a result consistent with previous work relating larger pupil diameter to increased variability (Murphy et al., 2014 [↗](#)) and individual differences in the temporal dynamics of evidence accumulation (Keung et al., 2019 [↗](#)). These findings join a growing body of work showing that pupil-linked arousal contributes to adaptive mechanisms that shape behavior to match current environmental demands.

Our findings further suggest that these pupil-linked arousal mechanisms operate separately from sensory adaptation to support flexible evidence accumulation. Adding trial-wise MT neural activity or evoked pupil diameter each increased explanatory power relative to behavior-only models, demonstrating that both signals provided behaviorally relevant information (Fig. 7 [↗](#)). However, the fact that their relative contributions were uncorrelated across sessions and that evoked pupil responses were unrelated to MT activity suggests that sensory adaptation and pupil-linked arousal may shape behavior through distinct mechanisms at different processing stages. Collectively, these findings support a framework in which sensory adaptation provides a “bottom-up” mechanism that modulates evidence encoding based on recent stimulus statistics, whereas pupil-linked arousal provides a “top-down” signal that adjusts how that encoded evidence is accumulated over time according to expectations about environmental stability. Together, these complementary processes may jointly implement the context-dependent “leak” described by normative models of evidence accumulation (Glaze et al., 2015 [↗](#); Murphy et al., 2021 [↗](#); Fig. 1 [↗](#)), with variability in their relative contributions across sessions and animals suggesting flexible recruitment that may depend on factors such as training, experience, or strategy.

Our conclusions are tempered by several limitations. First, we do not know precisely where the context-dependent adaptation measured in MT originates. In principle, these effects could, at least in part, be inherited from upstream visual areas, including areas V1 (Born & Bradley, 2005 [↗](#); Churchland et al., 2005 [↗](#); Movshon & Newsome, 1996 [↗](#); Rodman et al., 1989 [↗](#)) and V2 (DeYoe & Van Essen, 1985 [↗](#)), where adaptation to motion is also known to occur. Second, our recordings sampled single units across sessions and thus did not capture population-level changes in MT activity. Adaptation can influence activity beyond single neurons, including changes in correlated variability within neural populations that can affect decision making (Solomon & Kohn, 2014 [↗](#); Whitmire & Stanley, 2016 [↗](#)). Moreover, correlated variability in cortex can covary with pupil size

and LC-NE activity (Joshi & Gold, 2022 [↗](#); Vinck et al., 2015 [↗](#)), providing another potential link between arousal and sensory encoding that merits further study. Third, downstream cortical (Ding & Gold, 2012 [↗](#); Kim & Shadlen, 1999 [↗](#); Shadlen & Newsome, 2001 [↗](#)) and subcortical (Ding & Gold, 2013 [↗](#); Horwitz & Newsome, 1999 [↗](#); Lovejoy & Krauzlis, 2010 [↗](#)) areas that read out MT activity to form the decision may also contribute to flexible adjustments to the process of evidence accumulation. Future work combining simultaneous recordings and/or causal manipulations across sensory, decision, and arousal-related regions will be essential to disentangle how multiple distributed mechanisms implement adaptive evidence accumulation to support flexible decision making.

In summary, this work contributes to a broader reconceptualization of classic, hierarchical views of sensorimotor pathways that cast primary sensory cortex as a static encoder supplying input to flexible downstream decision processes that guide behavior. Instead, a growing body of evidence suggests that these early sensory areas can provide adaptive, context-dependent signals shaped by recent experience and environmental structure that can impact behavior (Waiblinger et al., 2022 [↗](#), 2025 [↗](#)). Our demonstration of separable, behaviorally relevant contributions from sensory adaptation in MT and pupil-linked arousal supports this view, indicating that flexible decision making in dynamic environments emerges from distributed mechanisms that include intrinsically adaptive signals in early sensory cortex that complement adjustments in downstream processes.

Materials and Methods

Three adult male rhesus monkeys (*Macaca mulatta*) participated in this study. Two of the monkeys (An and Ch) had extensive prior training on a standard (i.e., non-switching) version of the random-dot motion direction-discrimination task. All training, surgery, and experimental procedures conformed to the National Institutes of Health Guide for the Care and Use of Laboratory Animals and were approved by the University of Pennsylvania Institutional Animal Care and Use Committee (protocol #806027).

Behavioral task

The behavioral task (Fig. 1A [↗](#)) combined a change-point variant of the random-dot motion task (Glaze et al., 2015 [↗](#)) with adaptation-test elements from traditional MT sensory adaptation experiments (Van Wezel & Britten, 2002 [↗](#)). Briefly, each trial consisted of two sequential epochs: an adapting-stimulus epoch (2400 ms) and a test-stimulus epoch (100–1200 ms, randomly sampled from an exponential distribution with a mean of 500 ms). During the adapting epoch, a continuous random-dot motion stimulus was presented, within which motion direction switched at either a low (LSF; 1 switch) or high (HSF; 5 switches) frequency, creating two context-stability conditions. Change-point frequencies were designed so that total exposure time to each motion direction was matched across conditions (1200 ms per direction). Between epochs, there was a 50% probability of an additional change in motion direction, yielding an approximately equal number of switch and non-switch trials within each session. During the test epoch, the monkey maintained fixation and accumulated motion evidence over variable durations before reporting the final motion direction with a saccade to one of two choice targets. A juice reward was provided following correct choices.

For Monkey An (all 51 sessions), Mi (all 76 sessions), and Ch (11 out of 35 sessions), low (LSF) and high (HSF) switch-frequency conditions were run in blocks with counterbalanced order across sessions. The remaining sessions for Monkey Ch (n = 24) included interleaved LSF and HSF trials. Initial motion direction was counterbalanced across trials within each session. The adapting stimulus used high motion coherence (70%) to drive robust MT neural responses and adaptation, whereas the test stimulus used lower motion coherence (50–60%, adjusted across sessions to maintain overall task performance near ~70–80% correct) to promote temporal integration.

Eye position was monitored with a video-based system (EyeLink, SR Research) sampled at 1000 Hz and used to enforce fixation during motion viewing, register the saccadic response, and provide reward/error feedback online based on comparisons between the saccade endpoint and target

locations.

Analysis of behavioral data

Behavioral choice data were fit with a logistic function that modeled the probability of reporting a switch in motion direction as a function of test-stimulus duration and trial type (switch versus non-switch):

$$p(x) = \lambda + \frac{1-2\lambda}{1 + e^{-\beta_{slope}(x_1 - (\beta_{stay-switch}x_2 + \beta_{right-left}x_3))}} \quad (1)$$

where λ is the error (“lapse”) rate independent of motion information; β_{slope} governs the steepness of the psychometric function as a function of test-stimulus viewing time (x_1), signed by trial type (i.e., multiplied by 1 for switch trials and -1 for non-switch trials); $\beta_{stay-switch}$ captures biases related to reporting switch versus non-switch (x_2 ; switch = 1, stay = 0); and $\beta_{right-left}$ captures biases related to right versus left (x_3 ; right = 1, left = 0) choices. Here, “right” and “left” refer to all motion directions on the right or left side of vertical, respectively (e.g., motion directions between 270°–90° labeled “right,” and 90°–270° labeled “left”). Bias terms were encoded with respect to the x-intercept because the steep slopes can make the y-intercept difficult to estimate. We imposed an upper bound of 0.05 on β_{slope} , which captured the majority of variability across sessions. This formulation allowed for the construction of a time-dependent psychometric function in which β_{slope} reflects evidence accumulation over time. Low and high switch-frequency conditions were fit separately to allow for comparison of fitted slope parameters.

Electrophysiology

General surgical procedure and data acquisition methods have been described previous in detail (Law & Gold, 2008 [DOI](#)). Briefly, monkeys were prepared for experiments via surgical implantation of a head-holding device and recording cylinder. Area MT was targeted using stereotaxic coordinates, magnetic resonance imaging, and 3D reconstruction with BrainSight (Rogue Research, Inc). We searched for MT neurons with consistent spatial, direction, and speed tuning using a 99% coherence random-dot motion stimulus.

For each recording, multiple properties of the random-dot motion stimulus including aperture size and location, speed, and net motion direction were adjusted to match the tuning properties of the isolated single-unit and thus maximize stimulus-specific sensory adaptation. As such, exact stimulus parameters varied across sessions with stimulus sizes ranging from 4.6–15 degrees, locations within the contralateral visual hemifield ± 5 degrees from the point of fixation, speeds 2–8 degrees/s, and motion-directions distributed roughly evenly within circular space. The adapting stimulus began with preferred or anti-preferred motion chosen at random with equal probability and counterbalanced across trials.

Neural activity was recorded using polyamide-coated tungsten electrodes (FHC, Inc) or glass-coated tungsten electrodes (Alpha-Omega). Spike waveforms were stored and sorted offline using software from Plexon, Inc. In some sessions, multiple MT single units were recorded simultaneously (n = 40 units in 19 sessions).

Analysis of MT neural data

Firing rates were computed for each neuron and trial condition using a 100 ms sliding window advanced in 10 ms steps and aligned to the onset of the test stimulus. For each MT unit, firing rates were averaged across trial conditions, baseline subtracted on a trial-by-trial basis using responses from the 100 ms window immediately preceding adapting-stimulus onset as baseline, and normalized by the maximum firing rate observed during the 2400 ms adapting stimulus. Only correct trials were included in analyses unless otherwise indicated.

Direction selectivity was quantified as the average difference between responses to preferred and anti-preferred (null) motion from 100 to 500 ms after adapting stimulus onset, corresponding to the overlapping stimulus period (400 ms) across switch-frequency conditions with a 100 ms delay.

To classify units as “adapting” versus “facilitating,” average baseline-subtracted and normalized neural activity was compared between the first presentation of preferred motion (200–400 ms following adapting stimulus onset) and preferred motion during the test stimulus (50–500 ms). Units were classified as adapting if responses decreased by >2.5% and as facilitating if responses increased by >2.5%, under both LSF and HSF conditions. We chose a liberal criterion for classification because we were interested in maintaining enough neurons in each group for statistical comparison of sensory adaptation as a function of preferred-motion stimulus presentation. Alternatively, using linear fits to peak preferred-motion responses and classifying neurons based on whether the fitted slope differed significantly from zero ($p < 0.05$), yielded 34 units (21.9%) with negative slopes classified as “adapting” and 4 units (0.3%) with positive slopes classified as “facilitating,” which are comparable to prior work in sensory cortical areas (Dobler et al., 2024; Kheradpezhohu et al., 2017; Seay et al., 2020).

Analysis of pupil data

For each session, pupil data were aligned to the onset of the adapting stimulus and preprocessed by: (1) linear interpolation of missing values, (2) applying a first-order 5 Hz low-pass Butterworth filter, (3) removing and linearly interpolating samples exceeding ± 2 standard deviations from the mean, (4) subtracting the session-averaged response time course, and (5) z-scoring. For evoked-pupil analyses, only correct trials were included. For baseline pupil analyses, all trials were included so as not to disrupt the session’s temporal order.

Monkey Ch was excluded from pupil analysis because many sessions used interleaved (versus blocked) context-stability conditions, which removed the ability to establish trial-by-trial expectations of stability. Two sessions for Monkey An were also excluded because the pupil data were not properly saved.

To quantify context-stability effects on evoked-pupil dynamics, we applied sliding-window linear regressions (100 ms window, 10 ms slide; Equation 2) to estimate differences in evoked pupil response between LSF and HSF trials (β_2) while controlling for baseline pupil (β_1).

$$y = \beta_0 + \beta_{baseline}x_1 + \beta_{cxt}x_2 \quad (2)$$

where, $\beta_{baseline}$ estimates influence of baseline pupil diameter (x_1 : 100 ms window preceding motion stimulus onset) and β_{cxt} , indexes context-stability condition (x_2 : 1 = LSF, 0 = HSF).

Quantification of neural and pupil contributions to behavior

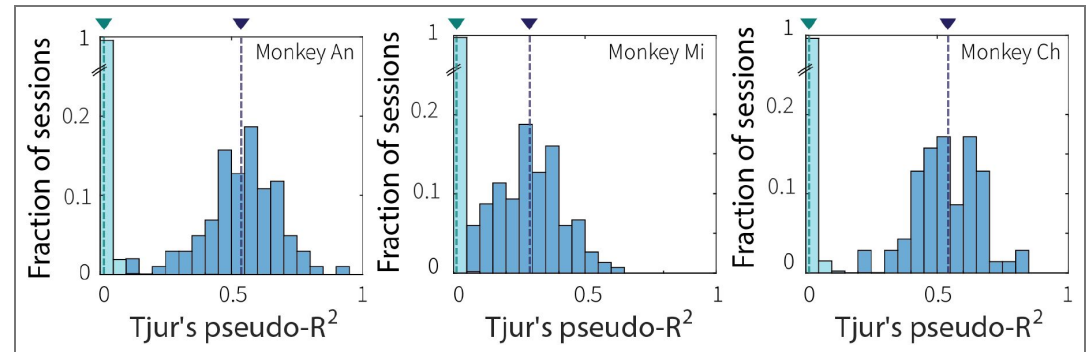
To quantify how neural and pupil measures contributed to behavioral sensitivity on a session-by-session basis, we added a term, $\beta_{trial-wise}$, to our time-dependent logistic function (Equation 3).

$$p(x) = \lambda + \frac{1-2\lambda}{1 + e^{-\beta_{slope}\beta_{trial-wise}(x_1*x_4 - (\beta_{stay-switch}x_2 + \beta_{right-left}x_3))}} \quad (3)$$

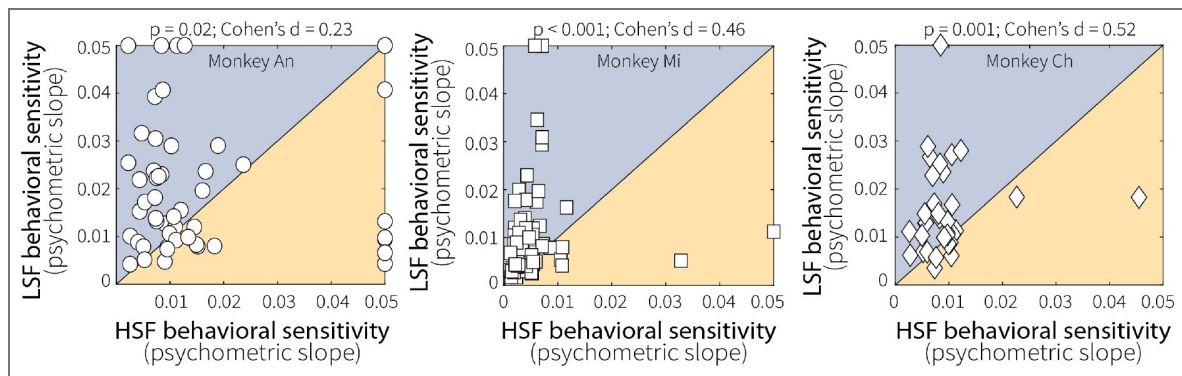
where, $\beta_{trial-wise}$ estimates the contribution of trial-wise MT activity for neural fits or trial-wise evoked pupil diameter for pupil fits (x_4).

For these fits, MT activity was computed as the average firing rate during the test stimulus on each trial with a 50 ms delay. Because MT neurons exhibit minimal response to null motion, only trials ending in preferred motion were used for neural fits. Evoked pupil diameter was calculated as the average of the residuals (500 ms window preceding test-stimulus onset) from a linear fit with average pupil diameter and baseline pupil diameter. Otherwise, parameterization of the fits was as described in Equation 1. Given the lack of evidence for an interaction between pupil-linked arousal and sensory adaptation, neural and pupil terms were assessed as the sole contributor in separate fits. When assessing explanatory power, for each session, the reported pseudo- R^2 value was the average across pseudo- R^2 values from separate fits to low (LSF) and high (HSF) switch-frequency trials.

Supplementary figures

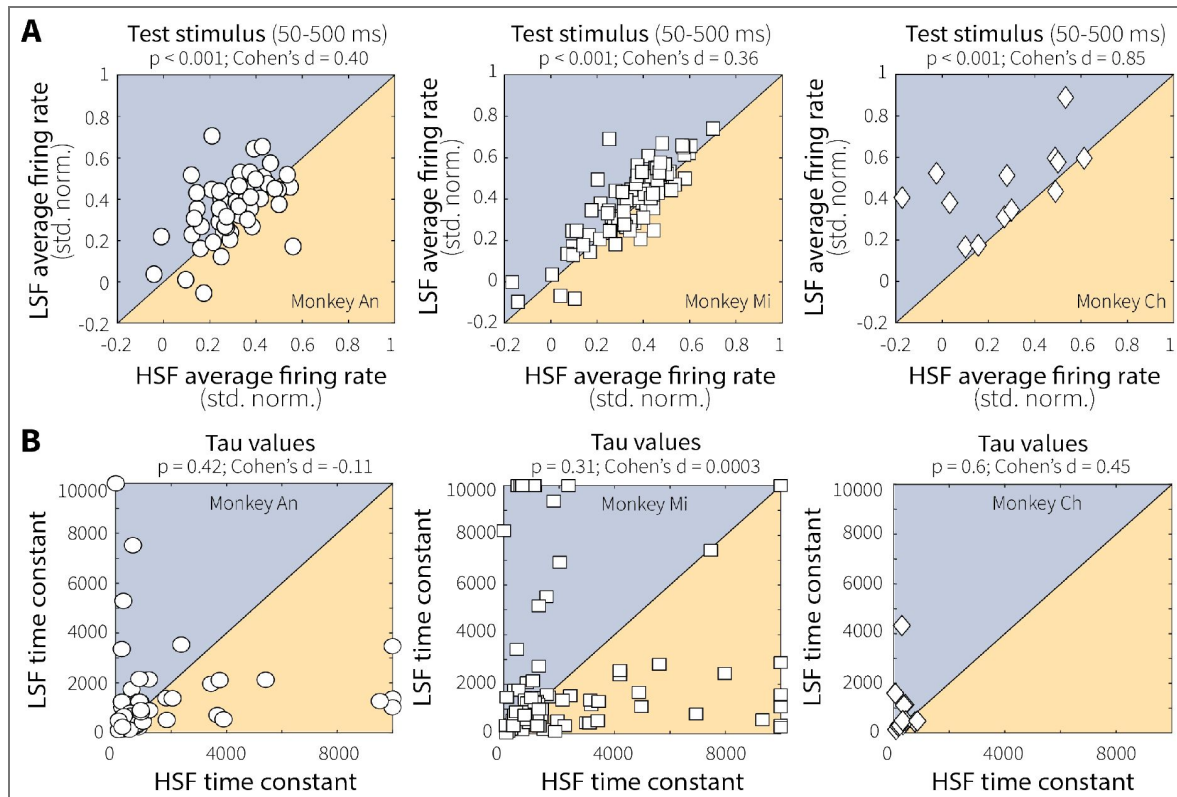


Extended Data Fig. 1. Time-dependent logistic model captures choice behavior. Distribution of Tjur's pseudo- R^2 values from empirical logistic fits (blue) compared with values from shuffled control ("null") fits (cyan) for each monkey, as indicated. Null fits were obtained by shuffling the association between test-stimulus durations and switch/non-switch trial types across 100 iterations per session. For each session, the reported pseudo- R^2 value was the average from separate fits to low and high switch-frequency trials. Dashed lines and triangles indicate mean values for each distribution. Enhanced explanatory power of the time-dependent logistic model versus the shuffled null was consistent across the three monkeys (Wilcoxon rank-sum test for equal medians: Monkey An: $p < 0.001$; Monkey Mi: $p < 0.001$; Monkey Ch: $p < 0.001$).



Extended Data Fig. 2. Context-dependent differences in evidence accumulation were consistent across animals.

Pairwise comparisons of fitted psychometric slopes for low (LSF) and high (HSF) switch-frequency conditions across sessions for each monkey, as indicated. Slope values were bounded at 0.05, which is roughly the maximum resolvable steepness given our sampling of viewing times. All three monkeys tended to have shallower time-dependent psychometric slopes at HSF relative to LSF (p -values are from a Wilcoxon signed-rank test of equal medians). This result implies reduced sensitivity to evidence as a function of viewing time, which is consistent with leakier evidence accumulation at HSF.

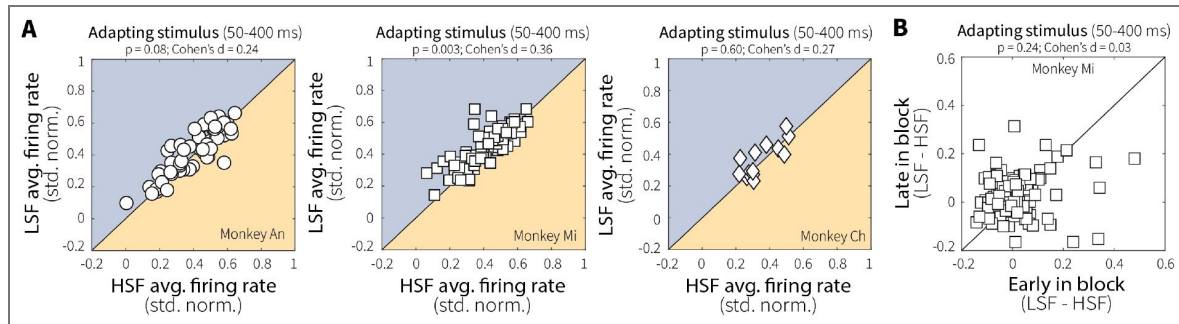


Extended Data Fig. 3. Context-dependent differences in evidence encoding were consistent across animals and reflected changes in response magnitude rather than temporal dynamics.

(A) Pairwise comparisons of mean responses of individual MT units (points) to preferred motion during the test stimulus (50–500 ms after onset) for low (LSF) and high (HSF) switch-frequency conditions across sessions for each monkey, as indicated. All three monkeys exhibited stronger MT responses at LSF relative to HSF (p -values are from a Wilcoxon signed-rank test for equal medians), demonstrating consistent, context-stability-dependent differences in evidence encoding. (B) We quantified the time course of MT neural response by fitting baseline-subtracted, normalized activity during the test stimulus (preferred-motion switch trials only) with a single-exponential function. To facilitate fitting, data for each neuron were divided into three bins (200–330 ms, 340–470 ms, and 480–600 ms following test-stimulus onset) chosen to accommodate initial response latency and include approximately equal trial counts. Fitting was performed separately for low (LSF) and high (HSF) switch-frequency conditions. The exponential time constant (τ , in ms) from best-fitting, single-exponential fits showed no difference between LSF and HSF conditions for each monkey (p -values are from a Wilcoxon signed-rank test for equal medians). Thus, the temporal dynamics of MT responses were not reliably affected by recent temporal statistics.

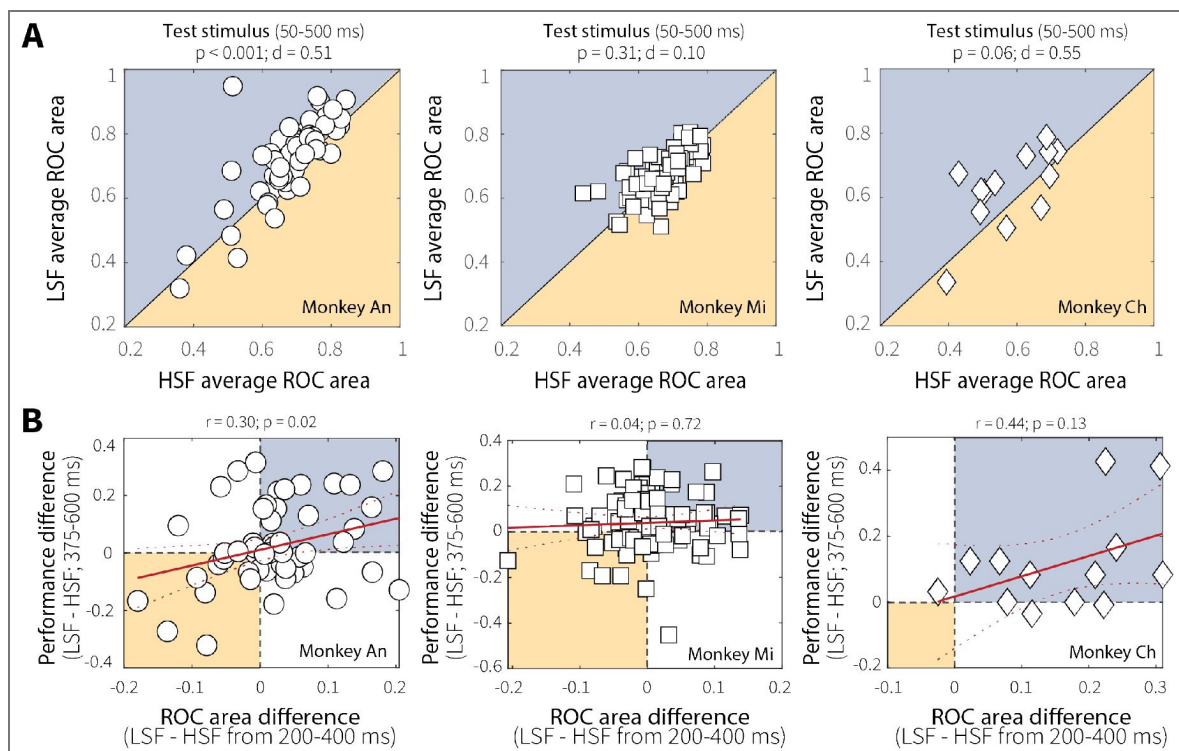
Extended Data Fig. 4. Context-dependent differences in initial MT responses were minimal and stable over the course of the session.

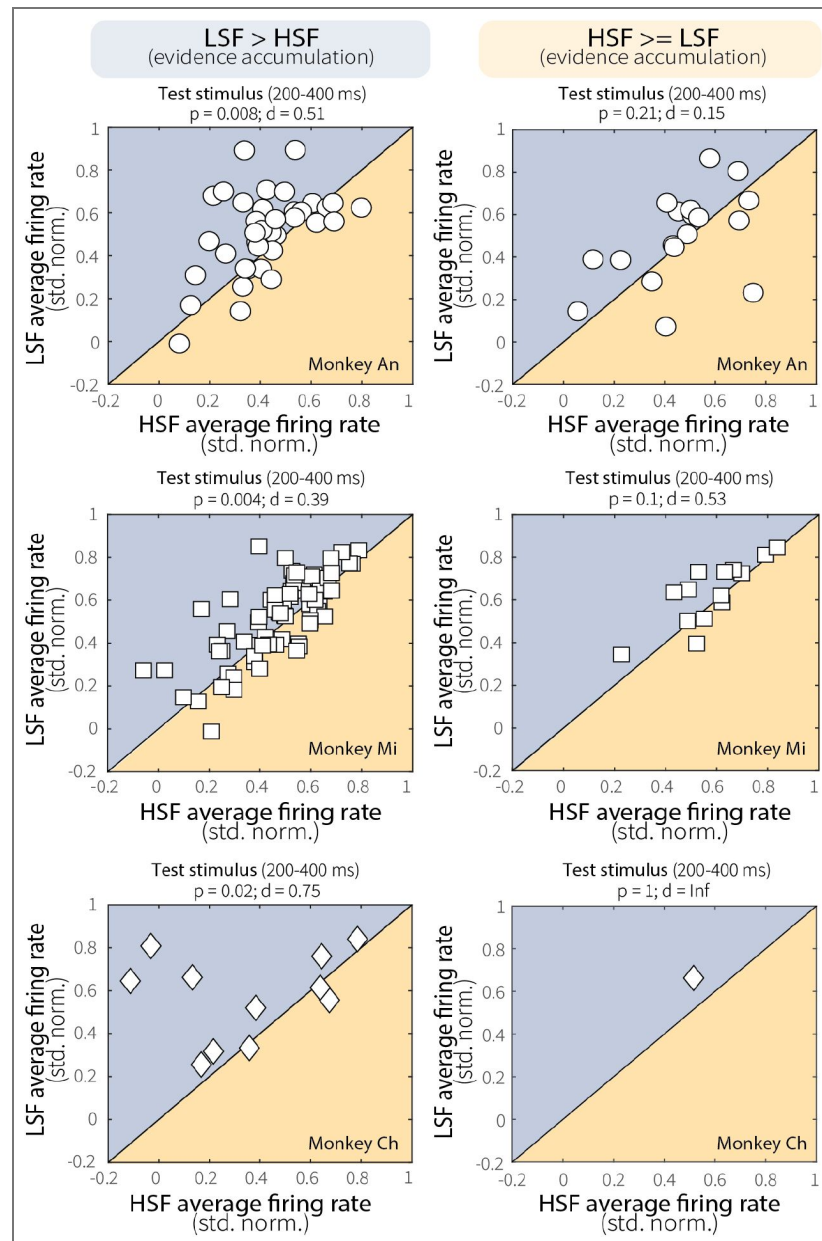
(A) Average MT single-unit responses (points) during onset of the adapting stimulus (50–400 ms) for low (LSF) versus high (HSF) switch-frequency conditions, shown separately for each monkey. Initial responses to preferred motion were matched between LSF and HSF for Monkey An and Monkey Ch, but were slightly elevated at LSF for Monkey Mi (p -values are from a Wilcoxon signed-rank test for equal medians). (B) For Monkey Mi, this initial difference in activity between LSF and HSF was stable across early and late blocks of each condition, possibly reflecting expectations about switching dynamics that were learned relatively quickly within each block of trials.



Extended Data Fig. 5. Context-dependent differences in MT evidence discriminability and its relationship to behavioral performance varied across monkeys.

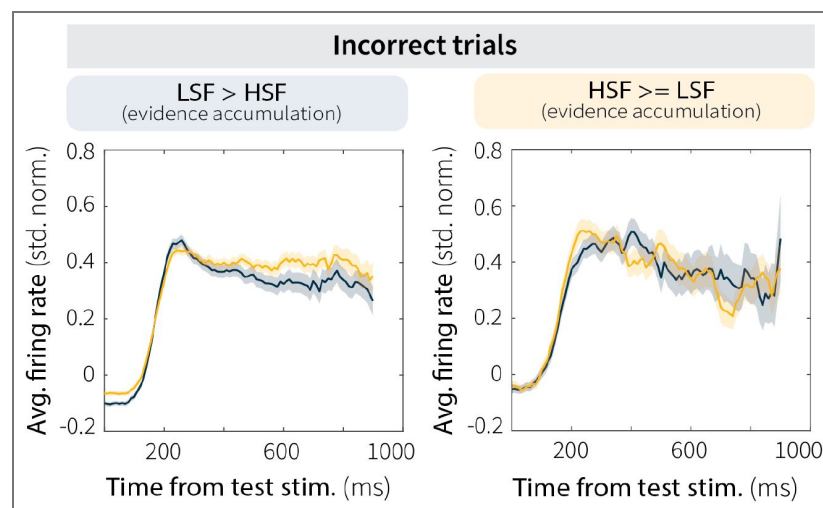
(A) Average ROC area during the stimulus (50–500 ms) for low (LSF) and high (HSF) switch-frequency conditions, shown separately for each monkey. ROC area was greater at LSF relative to HSF for Monkey An (Wilcoxon signed-rank test for equal medians). Monkey Ch showed a similar directional effect with a comparable effect size but for a much smaller sample and thus was not statistically significant. No effect was apparent for Monkey Mi despite a large sample size and context-stability differences in preferred-motion responses during the same stimulus window (Extended Data Fig. 3A). (B) Correlations between context-stability differences in ROC area (LSF–HSF; 200–400 ms) and behavioral performance (LSF–HSF percent correct; 375–600 ms), shown separately for each monkey. A significant positive relationship was observed for Monkey An, with a similar directional trend for Monkey Ch, though power was limited by the smaller number of recorded neurons. No relationship was observed for Monkey Mi despite a large sample size. Red solid and dashed lines indicate linear fits $\pm$ 95% CI.





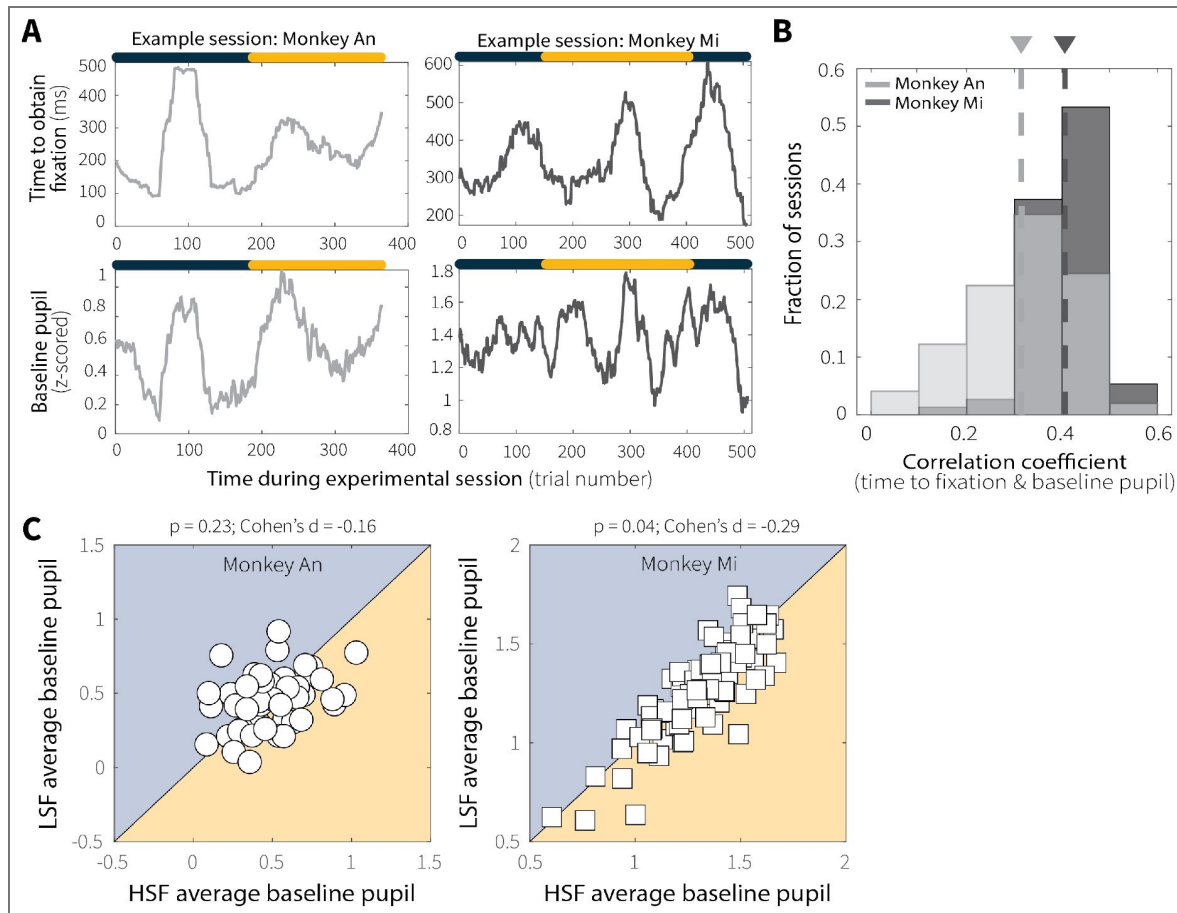
Extended Data Fig. 6. Relationship between MT neural activity and evidence-accumulation behavior was consistent across animals.

Average MT neural activity during the test stimulus (200–400 ms) for low (LSF) and high (HSF) switch-frequency conditions, shown separately for each monkey (rows) and session group (columns). Sessions were divided based on evidence-accumulation behavior: *left column*: sessions in which psychometric slopes were steeper at LSF than HSF ($LSF-HSF > 0$); *right column*: sessions in which psychometric slopes were greater at HSF or equal across conditions ($LSF-HSF \leq 0$). When the monkeys were more sensitive at LSF (left column), MT neurons showed greater activity at LSF relative to HSF. Conversely, when the monkeys were more sensitive at HSF or equally sensitive between conditions (right column), neural responses did not differ between conditions. *p*-values are for a Wilcoxon signed-rank tests for equal medians.



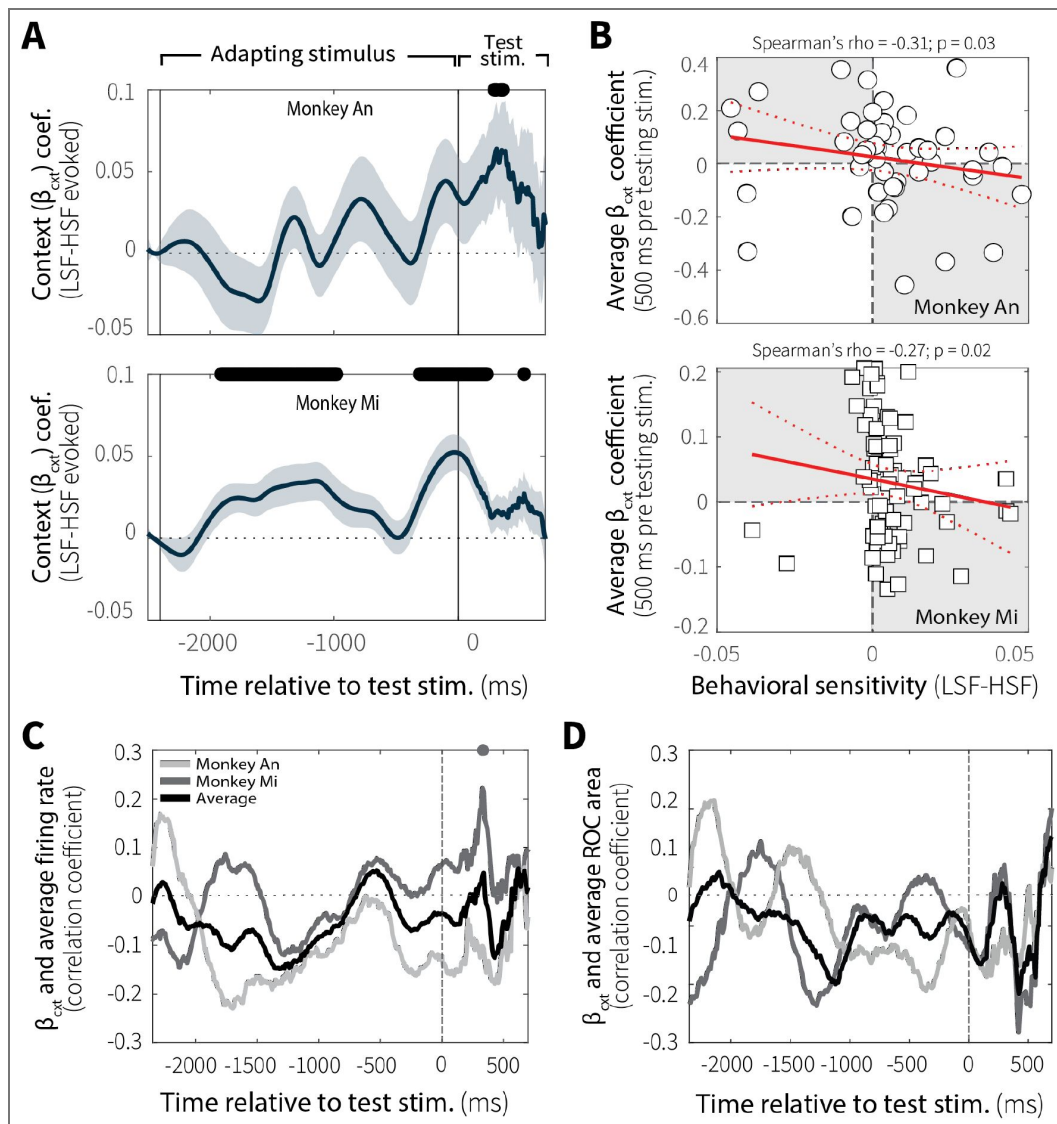
Extended Data Fig. 7. Context-dependent differences in MT activity were specific to correct trials.

Baseline-subtracted and normalized MT firing rates during the test stimulus for low (LSF; blue) and high (HSF; orange) switch-frequency conditions, averaged across incorrect trials for all sessions. Data are shown separately for sessions in which the monkeys were more sensitive (i.e., accumulated more evidence over time) at LSF (left; LSF $n = 1883$, HSF $n = 1939$) or HSF (right; LSF $n = 463$, HSF $n = 395$). Unlike correct trials, error trials showed no consistent modulation by context stability or evidence-accumulation behavior, further confirming that context-dependent changes in MT neural activity were behaviorally relevant.



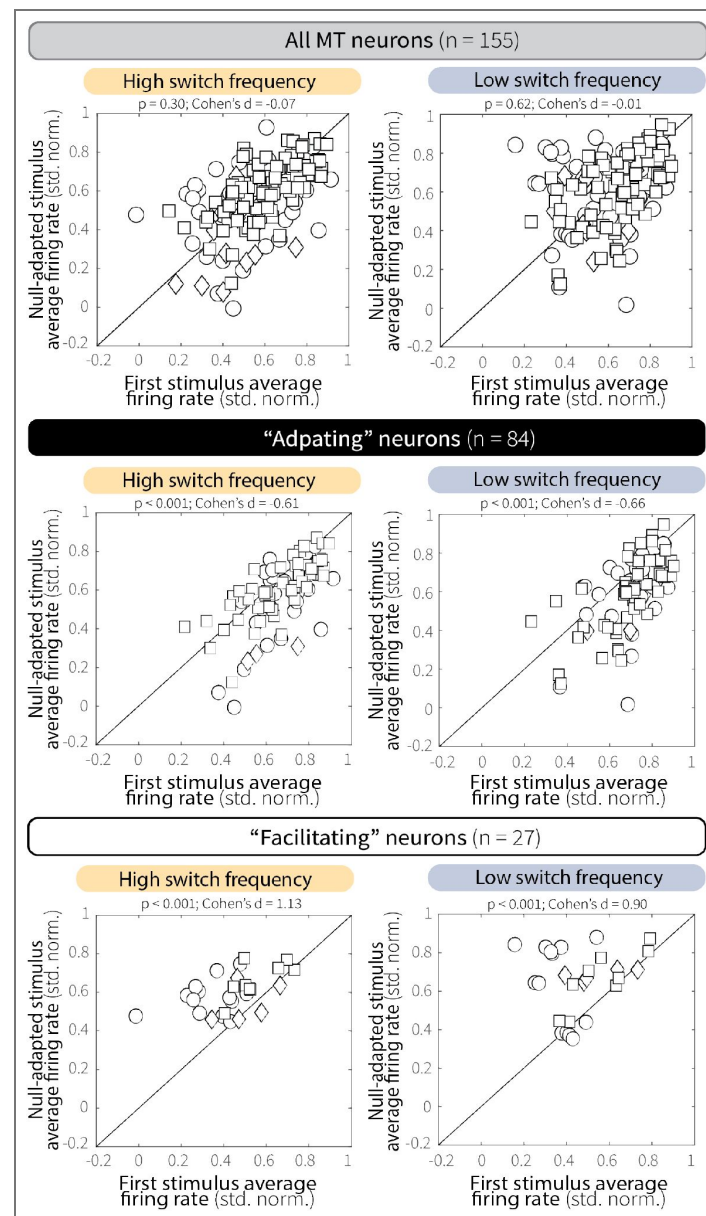
Extended Data Fig. 8. Baseline pupil diameter covaried with fixation-acquisition time across trials, sessions, and monkeys but did not differ across context-stability conditions.

(A) Representative sessions illustrating trial-by-trial relationships between fixation-acquisition time (top) and baseline pupil diameter (bottom) for Monkey An (left, grey) and Monkey Mi (right, black). The colored bar above indicates the active context-stability condition on each trial. (B) Distributions of session-wise Spearman's rank correlation coefficients between fixation-acquisition time and baseline pupil diameter across sessions for Monkey An (grey) and Monkey Mi (black). For both monkeys, the distribution of correlation coefficients was significantly positive (Monkey An: $p < 0.001$, $t = 18.6$, $df = 48$; Monkey Mi: $p < 0.001$, $t = 48.93$, $df = 74$). Mean correlation values are indicated by dashed lines and associated triangles. (C) Session-averaged baseline pupil diameter for LSF versus HSF conditions for Monkey An (left) and Monkey Mi (right). Average baseline pupil size did not reliably differ across context-stability conditions for monkey An and only slightly for monkey Mi (p values are for a Wilcoxon signed-rank test for equal medians).



Extended Data Fig. 9. Evoked pupil diameter related to context-dependent evidence accumulation-behavior, but not MT neural activity.

(A) Context-stability effects on evoked pupil diameter for individual monkeys. Sliding-window regression coefficients (β_{cxt} ; mean $\pm$ SEM across sessions) estimating differences in pupil diameter between low (LSF) and high (HSF) switch-frequency conditions while controlling for baseline pupil size. Black bar (top) indicates windows in which β_{cxt} differed significantly from zero ($p < 0.05$, uncorrected for multiple comparisons). (B) Average evoked pupil differences (β_{cxt} , -500–0 ms relative to test-stimulus onset) plotted versus behavioral sensitivity differences (LSF–HSF psychometric slope) from individual sessions for Monkey An (top) and Monkey Mi (bottom). Red solid and dashed lines indicate a linear fit $\pm$ 95% CI. (C) Spearman's rank correlation between context-stability differences in MT neural activity (LSF–HSF, 50–500 ms) and pupil β_{cxt} coefficients from individual sessions as a function of time relative to test-stimulus onset (computed in 100 ms bins with a 10 ms slide). Individual animals are shown separately (Monkey An: light grey; Monkey Mi: dark grey) along with the across-animal average (black). Corresponding colored bars indicate time points with significant correlations ($p < 0.05$, uncorrected for multiple comparisons). (D) Same analysis for MT discriminability (LSF–HSF ROC area, 50–500 ms) and pupil β_{cxt} coefficients as a function of time relative to test-stimulus.



Extended Data Fig. 10. Effects of null-motion adaptation on preferred-motion responses in MT.

Pairwise comparisons of MT responses to preferred motion before (first stimulus; solid orange line in Fig. 3C) versus after null motion (second, null-adapted stimulus; dashed orange line in Fig. 3C), shown separately for high (HSF, left) and low (LSF, right) switch-frequency conditions. Across the population (top row), prior exposure to null motion did not reliably alter subsequent preferred-motion responses. However, null-motion adaptation produced opposite effects in subgroups of adapting and facilitating neurons. Adapting single units (middle row) showed modest decreases in response to preferred motion following null-motion exposure, whereas facilitating units (bottom row) showed increased responses. Notably, the reduction observed in adapting single units was smaller than that following repeated preferred-motion stimulation. p-values are from Wilcoxon signed-rank tests for equal medians.

Data availability

All data and code have been uploaded to Box and are available at:

<https://upenn.box.com/s/7dx9enb1e1nx19tszrqaur8wrtrflf4>

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Additional information

Author contributions

Conceptualization, KDM and JIG; methodology, KDM and JIG; investigation, KDM; visualization, KDM and JIG; funding acquisition, KDM and JIG; project administration, KDM; supervision, JIG; writing – original draft, KDM; writing – review and editing, KDM and JIG.

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Peer reviews

Reviewer #1 (Public review):

Summary:

Effective decision-making in dynamic environments requires the brain to flexibly adjust how sensory evidence is accumulated over time, a process often modeled as an adaptive "leak." McGaughey and Gold propose that this flexibility is not solely a property of downstream integrators but is also supported by stimulus-specific sensory adaptation in the middle temporal area (MT). By recording single-unit activity in rhesus macaques during a motion direction-discrimination task, the authors found that more rapidly changing environments lead to reduced sensory encoding and discriminability in MT, which they argue accounts partially for a "leakier" integration. Furthermore, the study identifies pupil-linked arousal as a parallel, independent mechanism contributing to this adaptive process.

Strengths:

The study addresses an important question in cognitive neuroscience by exploring the neural substrates of perceptual flexibility. A major strength is the novel focus on how sensory adaptation, rather than just downstream integration, contributes to behavioral changes in dynamic environments. By shifting the perspective toward the encoding stage, the authors provide a more comprehensive account of how the brain manages evidence accumulation. This conceptual advance is supported by a rigorous experimental approach that combines human-like psychophysics with large-scale single-unit recordings in the middle temporal area (MT) and pupillometry.

Weaknesses:

(1) Alternative mechanisms for performance differences

The authors assume that the difference in performance between the low-switch (LS) and high-switch (HS) frequency conditions is explained by a change in the "leakiness" of integration. However, several other mechanisms could potentially explain this effect:

(i) Temporal Uncertainty: Integration might start later in the HS condition, leading to lower performance.

(ii) Reduced Efficiency: Integration could be less efficient in the HS condition (i.e., lower signal-to-noise ratio) without a change in the leak parameter itself.

(iii) Evidence Contamination: Motion information from the adapting stimulus in the HS condition may be integrated rather than ignored, which might be the case since the transition from the adapting to the test stimulus is not externally cued.

To distinguish between these alternatives, I suggest two possible analyses. First, a formal model comparison could be performed, though I acknowledge this may be inconclusive in the absence of response-time data. Second, an analysis of motion energy kernels could be revealing; the leak hypothesis makes the specific prediction that for long test stimuli, early samples should contribute more to the choice in the LS condition than in the HS condition, relative to late samples.

(2) Independence of neural and pupil-linked signals

The authors take the lack of session-wise correlation between context-dependent contributions from neural and pupil terms as evidence that these two signals provide independent contributions to the behavioral effect. However, could this lack of correlation simply be a result of high variability or noise in these estimates? The data shown in Figure 7B suggests that measurements are very noisy, which might obscure a potential relationship.

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Reviewer #2 (Public review):

McGaughey & Gold trained rhesus macaque monkeys to perform a motion-direction discrimination task in which a behaviorally irrelevant adapting stimulus with either fast or slow direction alternations preceded a variable-duration test stimulus, while simultaneously recording single-unit activity in area MT and pupil diameter. They report that adaptation to the more rapidly changing stimulus was associated with reduced behavioral sensitivity, attenuated test-evoked MT responses, and larger pupil-linked arousal signals. The authors interpret these behavioral changes as evidence for a more "leaky" evidence-accumulation process, and argue that this apparent leak is implemented in part through context-dependent sensory adaptation in MT and in part through arousal-related mechanisms. More broadly, they conclude that flexible evidence accumulation in dynamic environments arises from distributed adjustments across sensory encoding and neuromodulatory systems rather than solely from changes within a downstream accumulator. If correct, this interpretation has significant implications not only for our understanding of the neural mechanisms of perceptual decision-making but also for broader theories concerning the functional role of sensory adaptation.

The conclusions of the paper are mostly well supported by the data. Evidence for robust adaptation-induced changes in sensory encoding, behavior, and pupil dynamics is convincing, but further clarification and refinement are needed to establish a clear mechanistic link between these effects and decision-making processes.

Aspects of the behavioral analysis would benefit from a tighter connection between theoretical claims about evidence accumulation and the empirical features of the psychometric functions. For example, the rightward shifts observed across adapting conditions are interpreted as consistent with a reset of accumulation on switch trials, but similar patterns could also arise from failures to detect the test stimulus on a subset of trials, leading responses to default to the final adaptor direction. Likewise, changes in psychometric slope and asymptote are attributed to differences in evidence accumulation without explicit modelling or consideration of alternative explanations. Clarifying how specific features of the psychometric functions map onto distinct components of the decision process will strengthen the link between the theoretical framework and the behavioral data.

A slight concern is the lack of a consistent analytical approach for relating behavioral changes to neural and pupil-linked measures. Different sections of the manuscript rely on different behavioral metrics-such as differences in accuracy within a selected stimulus-duration range (e.g., Figure 5C) or psychometric slope differences (Figure 6C) - without clear justification for these choices. The analytical approach likewise varies between simple correlational analyses (Figure 5C, Figure 6C), pseudo-experimental group comparisons (Figures 5D, E), and the inclusion of neural or pupil terms in the behavioral psychometric regression model (Figure 7B). While each metric and approach may be defensible in isolation, adopting a more consistent framework will help convince readers that the reported effects are robust and not contingent on the selective choice of metric or analysis.

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Reviewer #3 (Public review):

Summary:

Environments change over time; therefore, optimal decision-making ought to discount older observations of the environment in favor of newer ones in a manner consistent with the amount of temporal instability. Computational models of perceptual decision-making model this temporal discounting with a 'leak' parameter that determines the rate at which older information is discarded. In this study, McGaughey and Gold examine the neurophysiological mechanisms that could underlie adaptation to different degrees of temporal instability. They developed a novel variant of the well-established perceptual decision-making random-dot-motion paradigm, in which the stimulus being evaluated was preceded by an 'adapting' stimulus with either high or low temporal stability. When the test stimulus was preceded by the adapting stimulus with lower temporal stability, NHPs showed reduced psychometric slopes, indicative of increased temporal discounting ('leak'). While the NHPs performed this task, single-unit neural activity was recorded in area MT, along with pupillometric data. The authors use these neural and pupil datasets to investigate two potential sources of adaptive discounting under varying amounts of temporal instability: sensory adaptation (changes in instantaneous evidence encoding), and arousal-related changes in evidence accumulation. MT neurons respond differently to the test stimulus under conditions of high vs low temporal stability of the adapting stimulus - when the adapting stimulus is more stable, MT neurons have larger and more selective responses to the test stimulus. In addition, evoked pupil responses to the test stimulus were modulated by the adapting stimulus. Both the strength of the difference in MT responses across contexts and the difference in pupil diameter across contexts were correlated with context-dependent modulation of the monkeys' behavior over sessions. The paper concludes that both sources appear to independently contribute to adaptive evidence accumulation, likely operating at different processing stages in the brain.

Strengths:

- (1) While computational models of perceptual decision-making have been very useful for

explaining behavior and neural responses in decision-making areas, we are still in search of some of the neural mechanisms that could implement such models. Studies such as this one, which aim to identify neural correlates of simplified model parameters, are quite crucial.

(2) Analysis is generally careful and well-executed.

(3) Prompts some interesting follow-up questions that could be answered with simultaneous recordings and causal manipulations, as the authors state in the Discussion - e.g., which areas are affected by arousal-related neuromodulation correlated with evoked pupil size and how.

Weaknesses:

(1) The task design may not be optimal. While the amount of time the monkey is exposed to each motion direction during the adapting stimulus is matched, it's hard to know if the reduced MT responses to the test stimulus are truly due to the greater frequency of switches during the HSF adapting stimulus or because the monkeys have been exposed to more repetitions of the stimulus. It's increased sensory adaptation in either case, but it makes it problematic to interpret this as temporal context-dependent adaptation specifically. I think this could potentially be partially addressed by an analysis that is in the paper, but could potentially be emphasized/fleshed out more, specifically the results shown in Figure 4D that seem to show that most of the reduction in neural response for adapting units occurs between the first and second stimuli.

(2) The pupillometric analysis seems to be an indirect way of assessing whether the accumulator itself might be modulated by temporal context, but the link could be made clearer. The authors show that context-dependent behavior is related to pupil size, which is related to arousal/neuromodulation, but it would be helpful to have some idea of what neural mechanisms underlying adaptive decision-making are actually impacted by this neuromodulation. Lacking neural data to address this question (e.g., from a brain region proposed to be involved in the accumulation process), at least more discussion of this would be helpful. Essentially, I'm unsure of how to interpret the pupil results: the argument that temporal context affects instantaneous evidence encoding in MT that then drives the accumulator is very clear, but I am a bit confused about what, mechanistically, I should think about the effect of neuromodulation doing.

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